

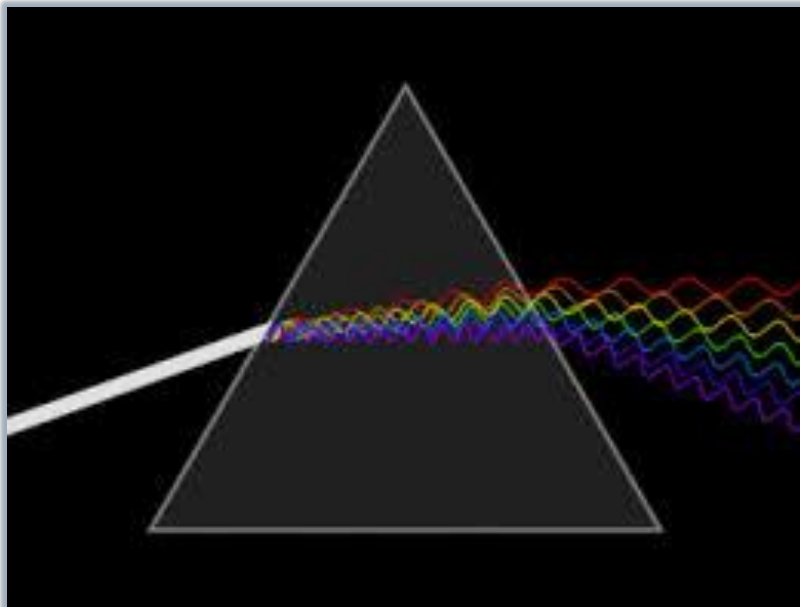


YEAR 9 SCIENCE

PHYSICAL SCIENCES

LIGHT AND HEAT

NAME: _____



WAVE MOTION

Waves transmit energy from one point to another in a substance (called a medium) without the particles of the substance moving with it.

This leads to the following definition.

Wave motion: a disturbance or vibration that transmits energy without the movement of matter from one point to another.

Waves that require a medium are called **mechanical waves**.

e.g. Sound, water waves, waves in strings and springs.

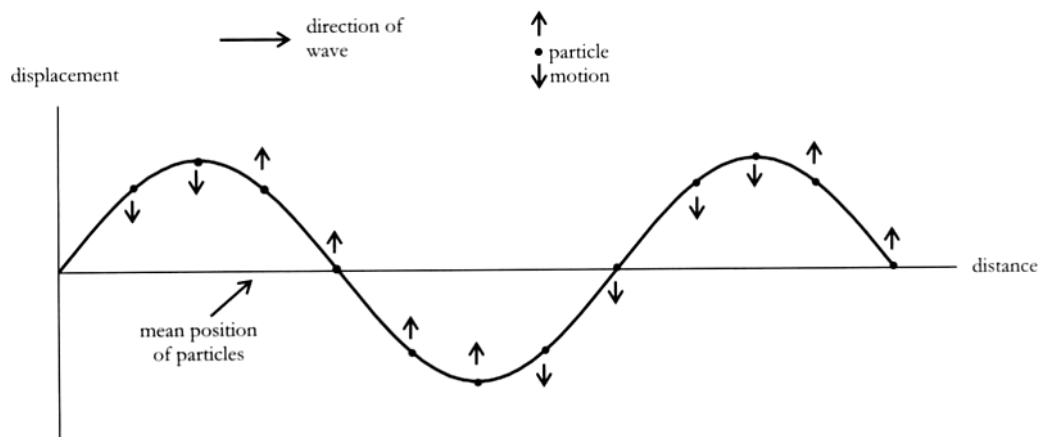
Waves that don't need a medium are called **electromagnetic waves**.

e.g. Light.

There are two types of mechanical waves.

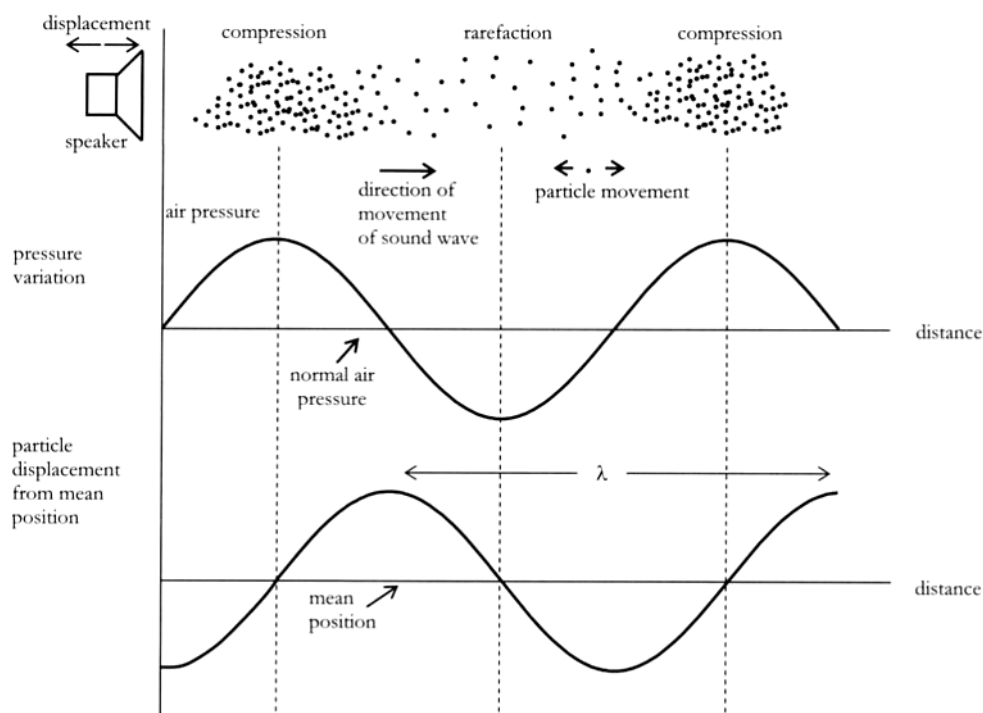
Transverse wave - particles move **perpendicular (up and down)** at right angles to the direction of the movement of the wave.

e.g. Water waves.



Longitudinal wave - particles move *parallel (back and forward)* in the same direction as the direction of the movement of the wave.

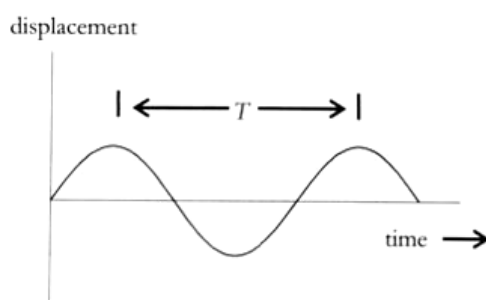
e.g. Sound waves



Wave graphs

Wave motion can be represented by a displacement/ time graph or a displacement/distance graph.

- Displacement / Time**

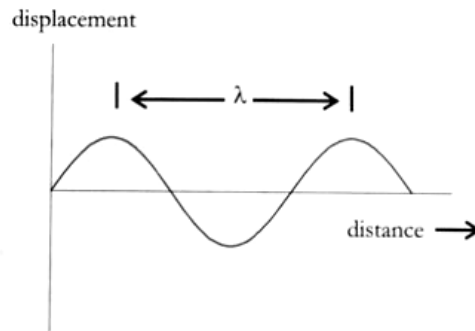


This graph represents the *motion of a single particle* as a wave passes through it.

The crests and troughs represent the particle's maximum displacement from its mean position.

T is the period of vibration - time taken for one complete oscillation (up-down and back to start position).

- **Displacement / Distance**

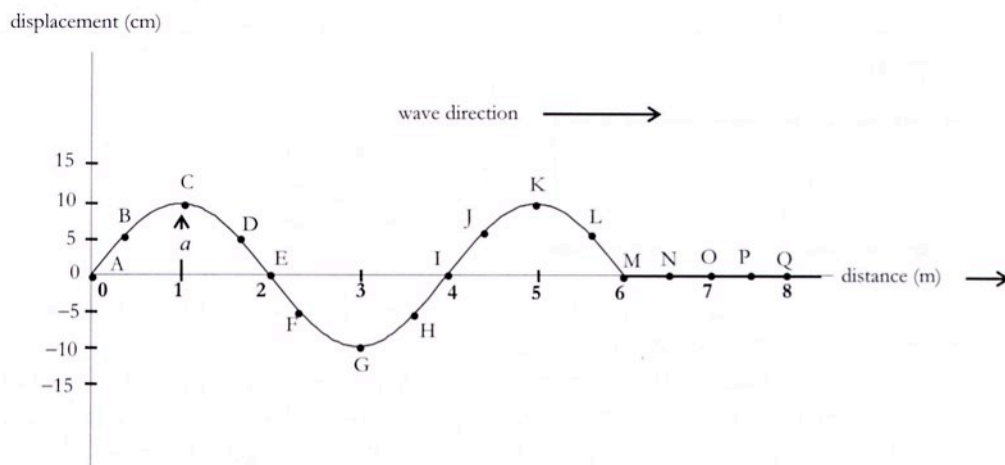


This graph represents the *movement of a number of particles* from their mean position at some instant in time.

Think of it as a "snapshot" or photograph that shows how the displacement of the particles of the medium varies over the distance at that instant.

λ represents the *wavelength of the wave*.

Describing Wave Motion - Important Terms



Wavelength (λ) The distance between any two crests or troughs.

e.g. The distance between points C & K, A & I, etc.

Amplitude (a) The maximum displacement of a particle from its mean (normal) position.

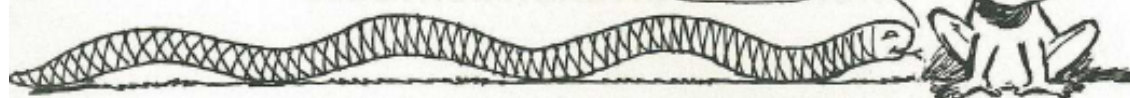
i.e. Height of a crest or depth of a trough.
The amplitude of the wave shown is 10 cm.

Frequency (f) The number of complete vibrations (or waves) per second.

The unit of frequency is the hertz (Hz).

Period (T) The time taken to complete one vibration (or to produce one complete wave).

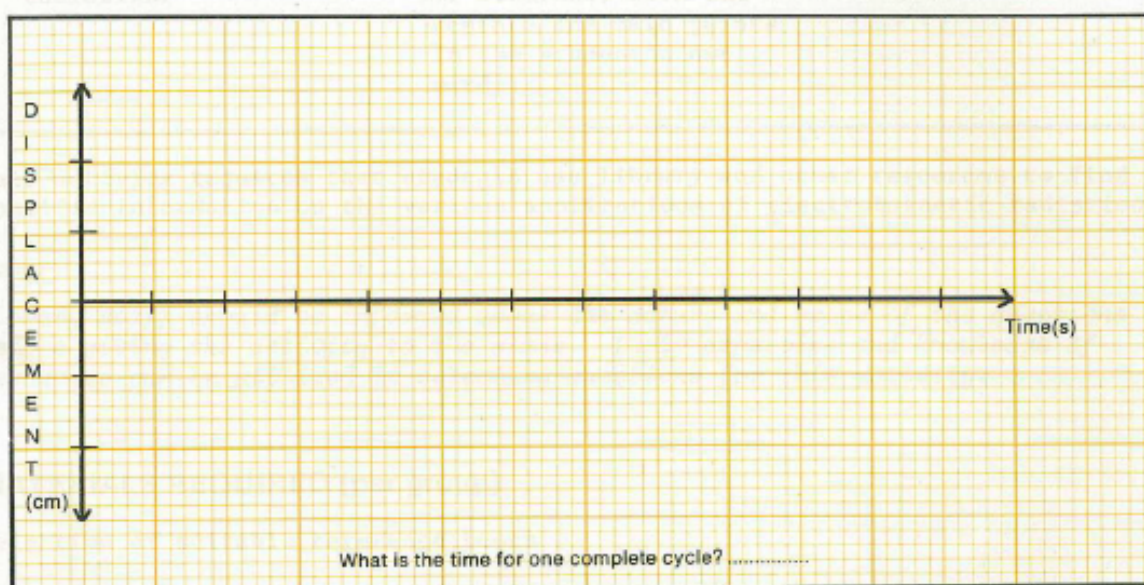
SKETCHING WAVES



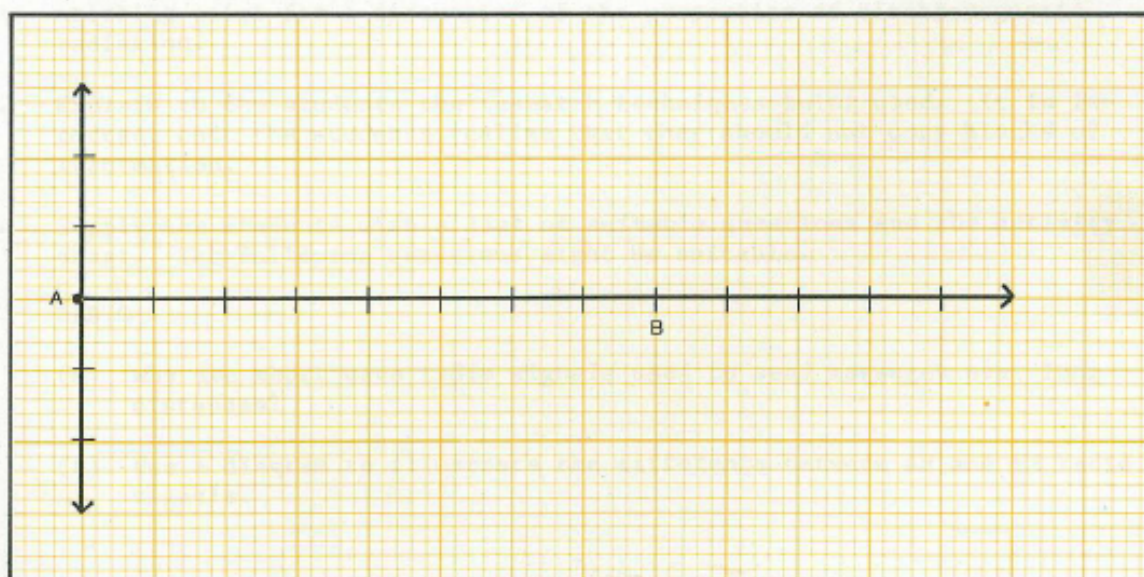
The following data have been obtained by photo-analysis of the motion of a single particle in a medium as energy in a wave form passes through the medium.

Time (s)	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
Displacement from rest position (cm)	0	5	11	16	19	20	19	16	11	5	0	-5	-11	-16	-19	-20	-19	-16	-11	-5	0

Plot the data on the graph paper.



Using the piece of wire you have been given, construct the outline of a wave with a wavelength of 4 cm and an amplitude of 3 cm. Trace the outline onto the graph paper.

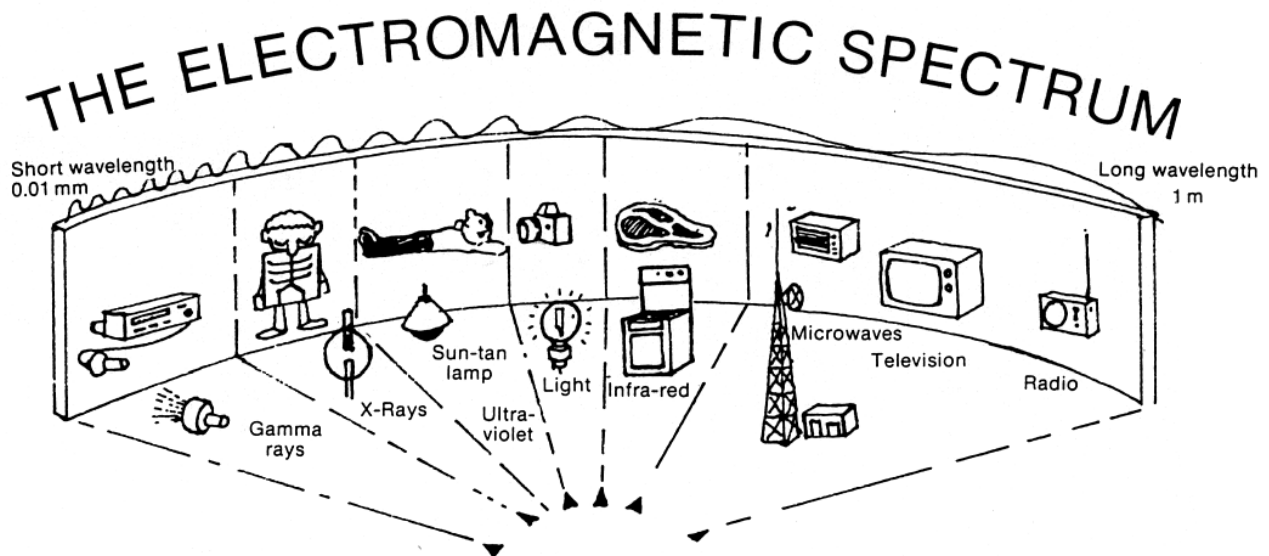


If the wave takes 1 second to travel from A to B, calculate the frequency.

ELECTROMAGNETIC SPECTRUM

Light is pure energy and is called *electromagnetic radiation* - it is made up of an electric and magnetic field, hence the name.

There are several different forms of this radiation, each with its own characteristics and use. This is shown below in the *electromagnetic spectrum*.



Power lines - moving current can generate a form of light.

Radio / TV - used for communication.

Microwaves - cook food by causing water molecules to vibrate, can be used for communication.

Infra-red - felt as heat.

Visible - we see in this region.

Ultra-violet - causes skin cancers, helps plants to grow.

X-rays - penetrates body well, used in medicine.

Gamma rays - very high-energy radiation given off during nuclear reactions or explosions.

**YEAR 9 SCIENCE
PHYSICAL SCIENCES
INVESTIGATION - THE SPEED OF LIGHT**

Aim: To determine the speed of light.

Equipment:

- Microwave oven
- Family block of chocolate
- Ruler
- Small ceramic bowl

Method:

1. Raise the turntable in the microwave by placing an upturned ceramic bowl over the rotator. **The turntable must not turn.**
2. Unwrap the chocolate bar and place it in the centre of the turntable with the **smooth side facing upwards**.
3. Start the microwave and watch the chocolate until evidence of melting occurs. This may be as the chocolate starts to discolour or may go "shiny".
4. Stop the microwave and allow the chocolate to cool.
5. Use a ruler to measure the distance between the "hot spots" that show on the chocolate. Record them in the table below and find the average.
6. Record the frequency of the microwave - it is on the plate at the back of the appliance. It is usually in gigahertz (GHz).

Background:

The speed of any wave is given by:

$$\text{speed} = \text{frequency} \times \text{wavelength}$$

The frequency will need to be converted to Hertz (Hz).

The wavelength needs to be in metres (m).

This will give the answer in metres/second (m/s).

The microwaves inside the oven bounce around and form what are called "standing waves". This means they have distinct points where the energy adds together, causing "hot spots".

The distance between adjacent "hot spots" is half of a wavelength.

Results:

Distance between "hot spots" (cm)

Average distance between "hot spots" = _____ cm

= _____ m

Wavelength = 2 x average distance = _____ m

Frequency: _____ GHz (multiply by 1,000,000,000)

= _____ Hz

Speed = frequency x wavelength

= _____ x _____

= _____ m/s

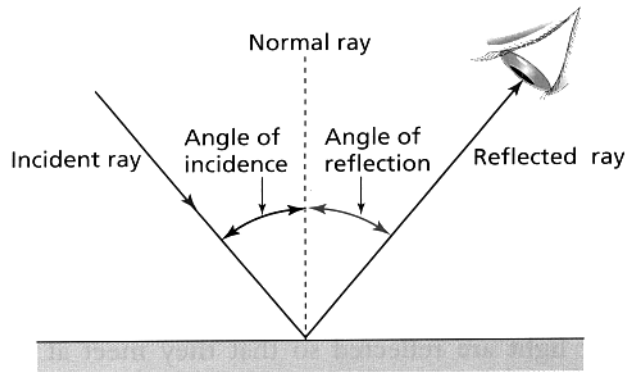
Conclusion:

The speed of light is _____ .

REFLECTION

Objects can be seen because light reflects from them into the eye. The rays of light follow the following rules.

LAWS OF REFLECTION

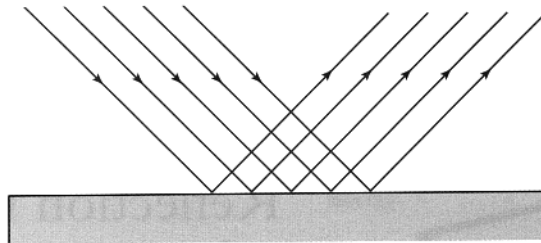


1. The angle of incidence = the angle of reflection.
2. The incident ray and reflected ray are in the same plane as, but on opposite sides of, the normal.

REGULAR AND DIFFUSE REFLECTION

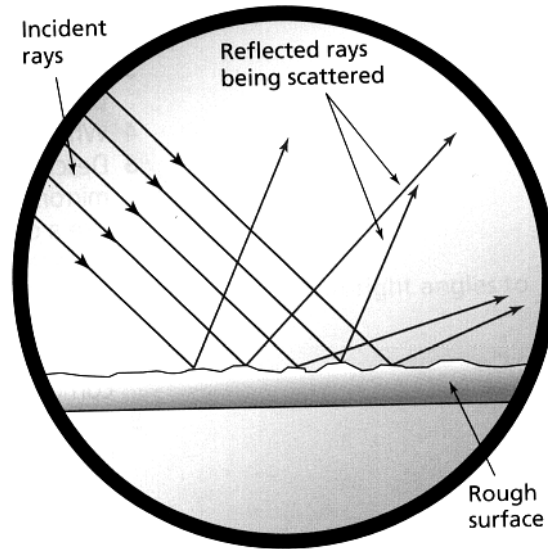
Regular reflection - parallel rays of light that hit a smooth surface are reflected parallel to each other.

e.g. mirror, polished silver



Diffuse reflection - parallel light rays hitting a rough surface are reflected in all directions (the light is scattered).

e.g. newspaper, sandpaper, "matt" paint on walls.



IMAGES FORMED IN MIRRORS

IMAGES 23QAMI IMAGES

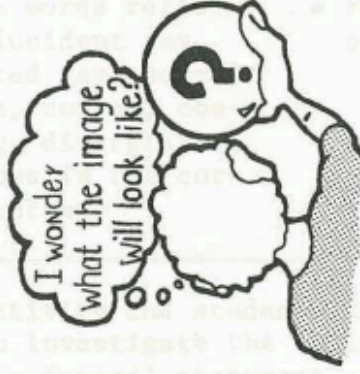
IMAGES? IMAGES

LOCATE THE IMAGE



1. Draw a line AB on a piece of paper. In the centre of AB draw the line CO. Measure the length of CO.
CO = ____ cm.
2. Draw in the other lines shown on the diagram and place a pin at point C.
3. Place a mirror on the line AB so that its reflecting surface is on the line.
4. Look into the mirror as shown and line up the image of one of the lines in the mirror by sighting along the edge of a ruler. When it is lined up, rule a sight line along the edge of the ruler. Repeat this for all of the other lines.
5. When all the sight lines have been drawn, remove the mirror and extend CO and all the other sight lines so that they meet. Label this point X.
6. Record your observations.
How do OC and OX compare?

MIRROR MIRROR



In this part of the activity you will be using three different types of mirrors.

Starting with the mirror at arm's length, slowly bring the mirror towards your eye.

Describe any changes in the images.

MIRROR

OBSERVATION - what does the image look like?

CONCAVE

CONVEX

Place a plane mirror above and at the ends of the words below.

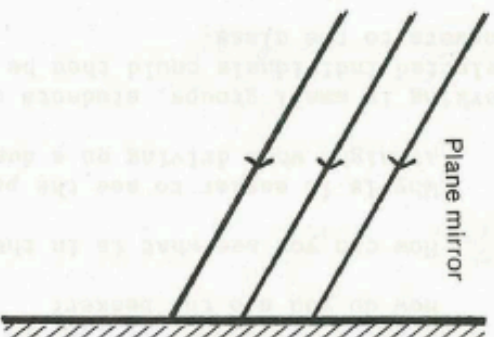
CARBON DIOXIDE

ECNALUBMA

Briefly describe what you observed.

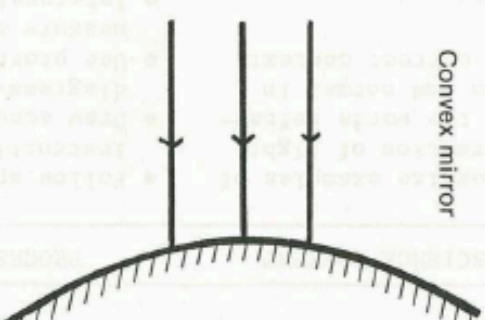
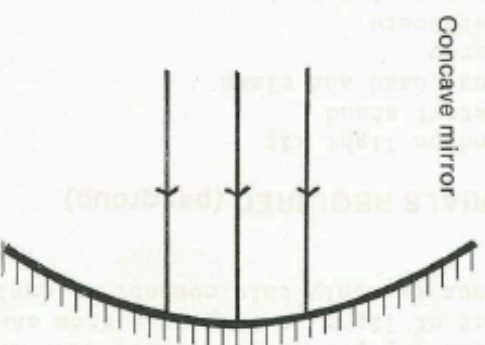
REFLECTION

MIRRORS

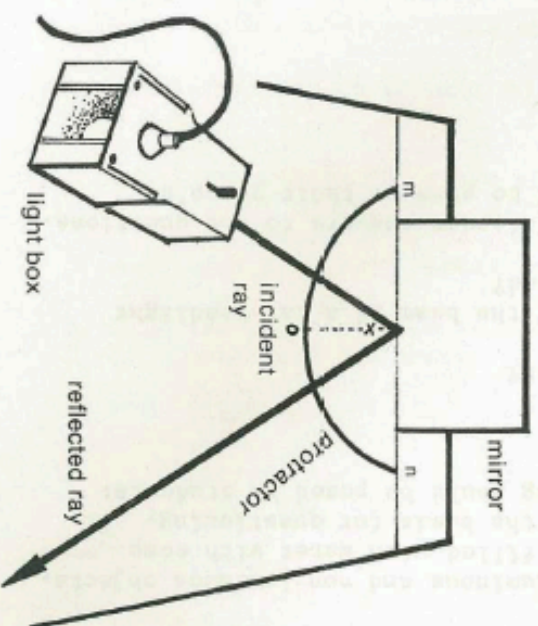


ANGLES

TRACE THE RAY



REFLECTION



MEASURE THE ANGLE

PROCEDURE

Using the prepared protractor, place the plane mirror so that its reflecting surface is on the line MN.

Using a single ray of light, start at O and position the light box at various angles between O and M. The angle between OX and the incident ray is called the incident angle and is recorded in a table of results. The angle of reflection (the angle between OX and the reflected ray) is also recorded.

RESULTS

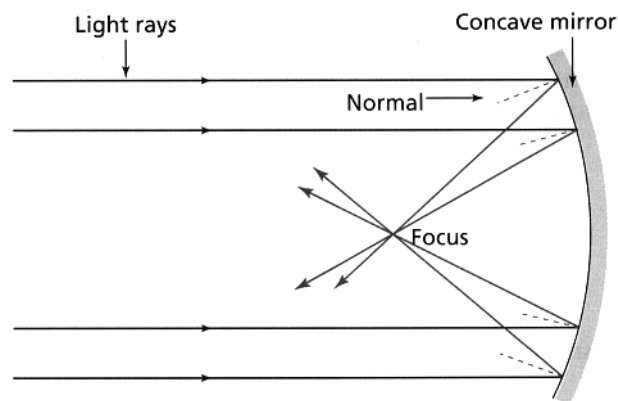
ANGLE OF INCIDENCE	ANGLE OF REFLECTION

CONCLUSION

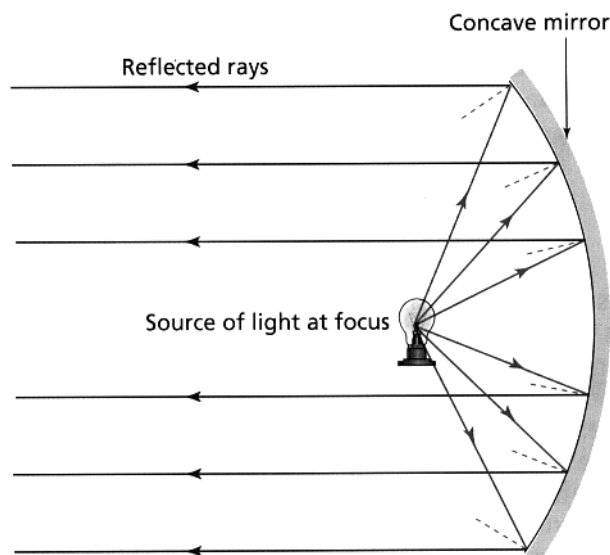
CURVED MIRRORS

Concave Mirrors

- Silvered surface curves *inwards*.
- Parallel rays of light *converge* to a point (called the *focus*).
- Used as make-up or shaving mirrors (magnifies the face), reflecting telescopes, solar ovens to reflect sunlight.
- Images produced vary from real to virtual, large to small, upright or upside down - depends on how far the object is from the mirror.

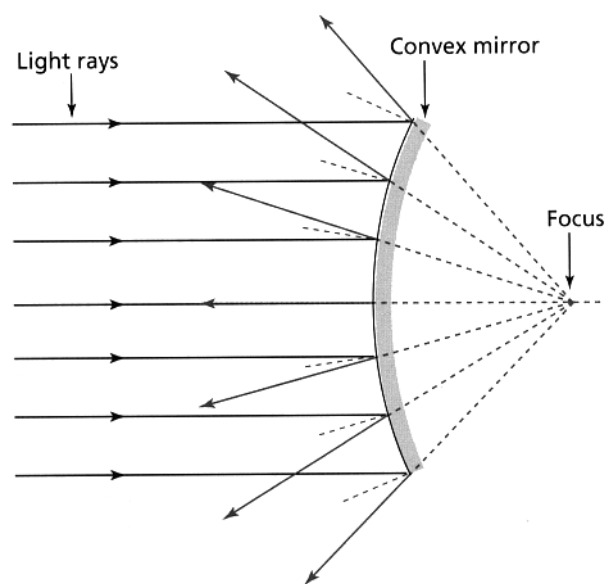


- A light source placed at the focus produces a beam of parallel light rays.
- Used in torches, car headlights, spotlights.



Convex Mirrors

- Silvered surface curves *outwards*.
- Parallel light rays *diverge* from a *focus point behind the mirror*.
- Used as security mirrors in shops and rear vision mirrors in cars.
- Give a wide field of view but make judging distance very difficult - everything looks smaller.
- Image is always *virtual, upright* and *smaller*.

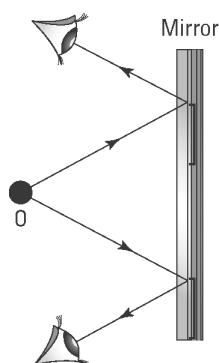


Reflection and scattering of light

Student: Class:

Locating the virtual image

1. An object (O) is observed by looking at the rays of light reflecting off a plane mirror. The mirror consists of a plate of glass with a silver back surface mounted on a sheet of wood.



- (a) Use a ruler and pencil to extend each reflected ray (using dotted lines) until they intersect behind the mirror. Label the point of intersection as 'virtual image'.
- (b) Complete the sentence:
The object in front of a reflecting surface and a virtual image are the distance from a mirror.
- (c) Draw the *normal* to the mirror surface and measure the incident and reflected angles for the ray above the object, O.

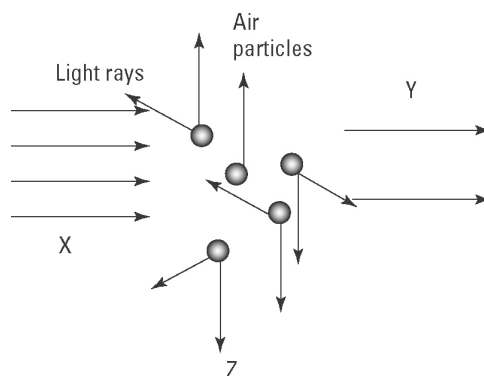
Light and air

2. The diagram below shows the collision of light rays with particles of air in the atmosphere. Use the letters X, Y and Z to identify the following rays.

Scattered:

Incident:

Transmitted:

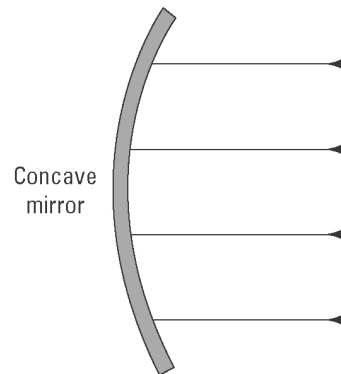
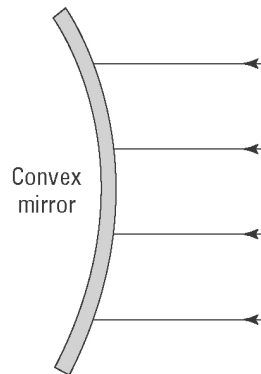


Curved mirrors

Student: Class:

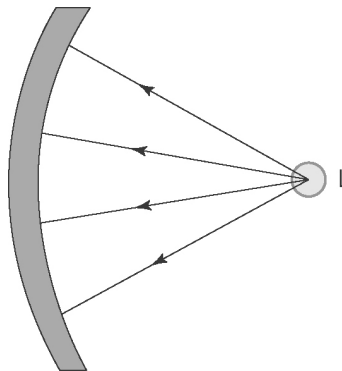
Reflection from convex and concave mirrors

- Four parallel light rays are incident on a convex and a concave mirror as shown. Complete the diagrams (using a pencil and ruler) to show the reflected rays.



Light emitted from the focal point

- The following diagram shows four light beams emitted from a light globe (L) placed at the focal point of the concave mirror.
 - Use a pencil and ruler to show the direction of the light rays reflected from the mirror.



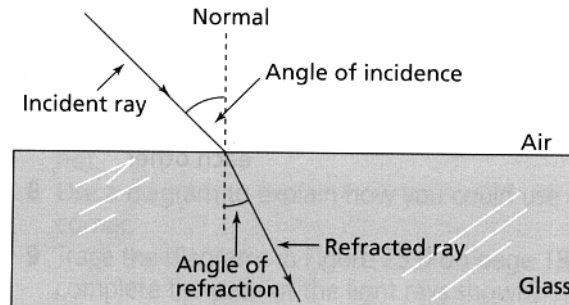
- Describe these reflected rays.

.....

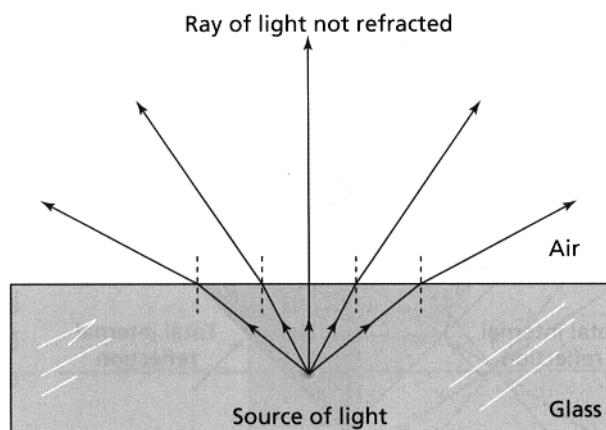
.....

REFRACTION

Refraction - bending of a light ray as it enters a new medium at an angle and its speed changes.



A ray of light entering an *optically more dense medium* (e.g. water, glass), it *slows down* and is *refracted towards the normal*.



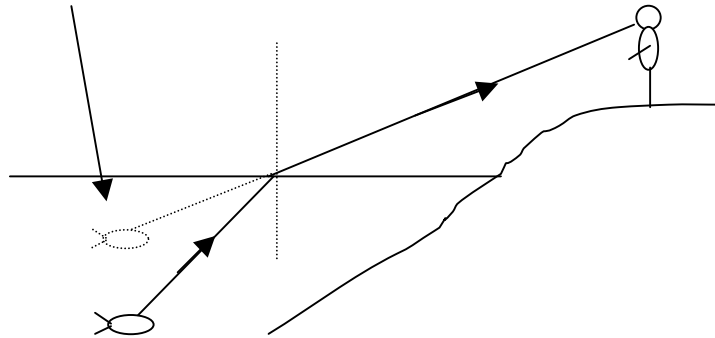
A ray entering an *optically less dense medium* *speeds up* and *refracts away from the normal*.

A ray that enters along the normal at 90° to the new medium will change speed but not direction.

Examples

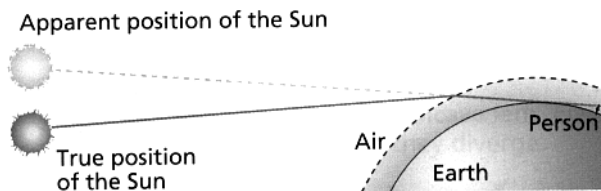
1. Why does a fish appear closer to the surface or a pool appear shallower than it really is?

Apparent position of fish is closer to the surface.



The light from the fish refracts away from the normal as it enters the air so the person thinks the fish is closer to the surface.

2. Why do we get about 4 minutes of extra sunlight each day?



The light from the Sun refracts down to the surface on entering the atmosphere.

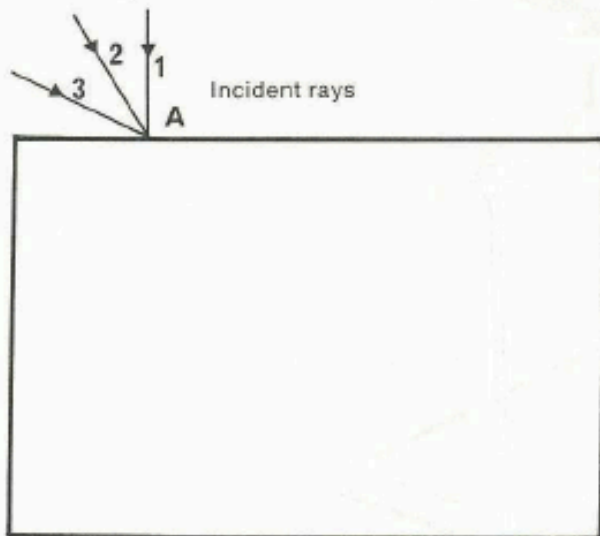
Hence we see the Sun after it has gone below the horizon.

REFRACTION

BENDING LIGHT RAYS

RAYS AND PRISMS

Using the prisms and the light box, shine a single ray on the paths shown on the diagrams. Draw the path of the rest of each ray on the diagrams.



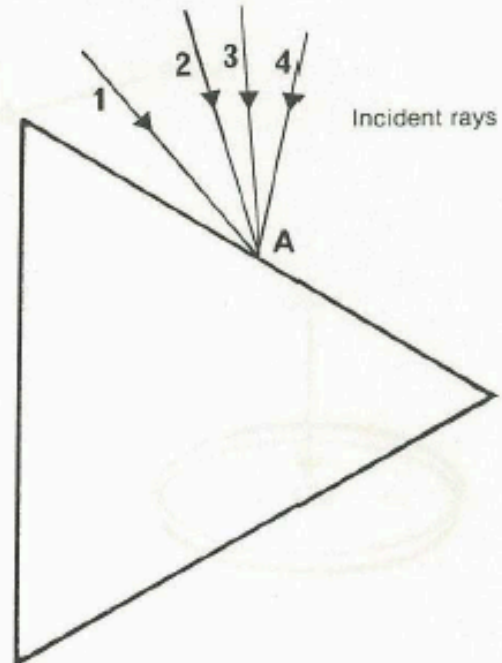
Measure the angle at which ray 1 enters the block.

What do you observe about the path of ray 1?

Do the incident and emergent rays line up?

What do you see when you look through the plastic block while the light is shining?

Do your ray paths look the same as the paths you have drawn?



Measure the angle at which ray 1 strikes the block.

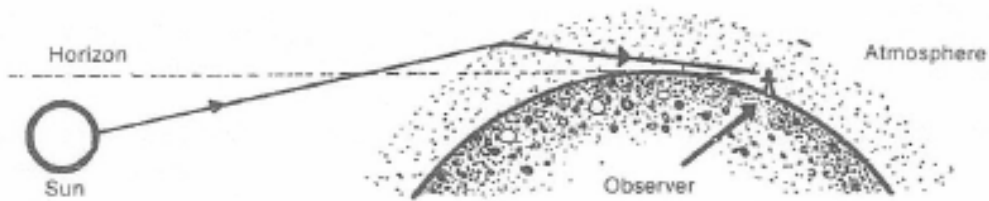
What do you observe about the path of ray 1?

Briefly describe what happened to incident rays 3 and 4.

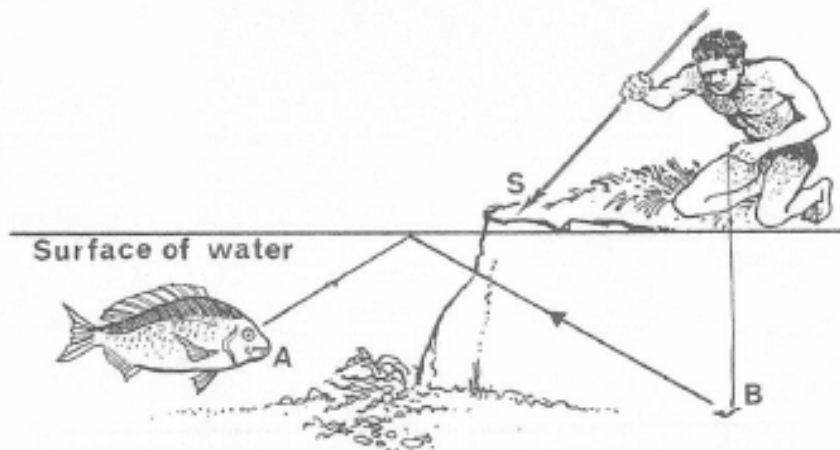
Do the incident and emergent rays line up?

APPLICATIONS OF REFRACTION

Explain why a person is able to see the sun when it is below the horizon. Where is the apparent position of the sun?



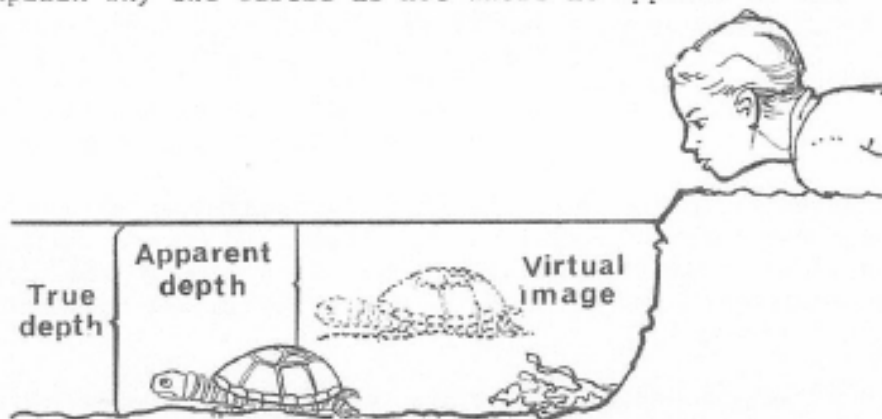
Briefly explain why the fish cannot see the man on the shore, yet the fish is able to see the worm.



Explain why the boat man appears to be rowing with broken oars.



Explain why the turtle is not where it appears to be.

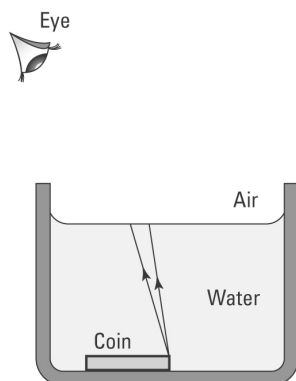


Refraction

Student: Class:

The appearing coin

1. A coin is placed in an aluminium beaker so that it is just out of sight when looking into the beaker from an angle, as shown. Water is then added to the beaker, so that when the water reaches a certain depth, the edge of the coin can be seen.



- (a) Use a ruler and pencil to show how the two light rays from the edge of the coin will enter the eye so that the edge becomes visible to the person when the water is added.

- (b) Explain why these rays bend as they enter the air.

.....

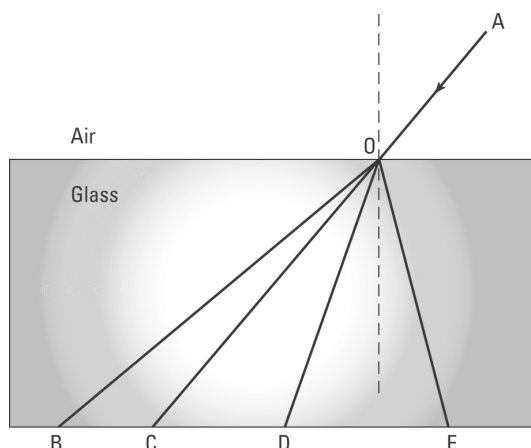
.....

.....

- (c) On the diagram sketch the apparent position of the coin and show why the coin appears to be at a more shallow depth.

Refraction of light through a rectangular prism

2. A light ray is incident on one side of a glass rectangular prism as shown.



- (a) The light ray travels from A to O. Identify which of the following paths it will take on entering the glass prism.

- A. OB
B. OC
C. OD
D. OE

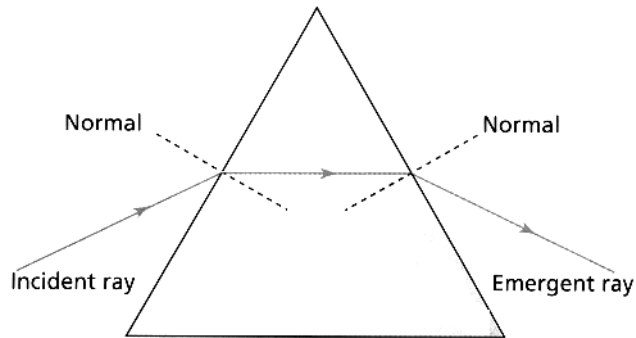
Answer:

.....

- (b) Use a pencil and ruler to show the path of the light ray as it emerges back into the air at the base of the prism.

LENSES

If a ray of light passes through a glass prism, it will change its direction as shown.

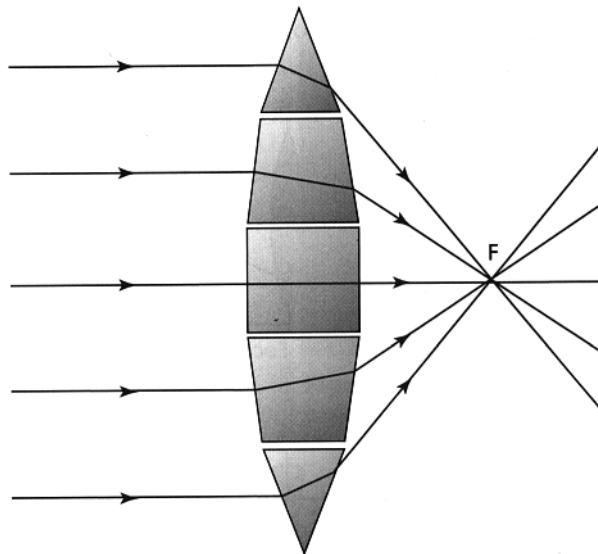


It is this behaviour that is behind the working of lenses.

There are two types of lenses.

1. **Convex** (Thicker at the centre.)

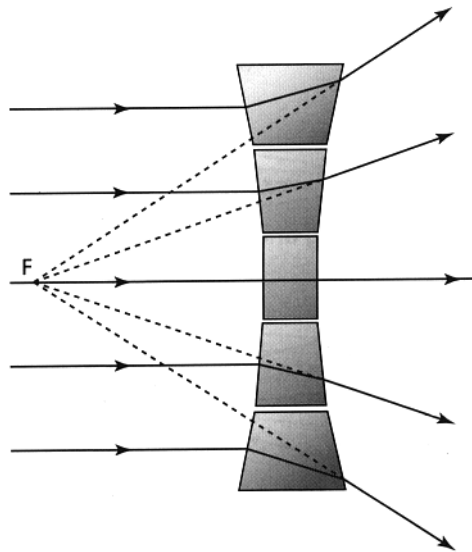
- Converges parallel rays of light to a point.



- Each part of the lens acts as a prism to change the direction of the light rays.
- Sometimes called a *converging lens*.

2. **Concave** (Thinner at the middle.)

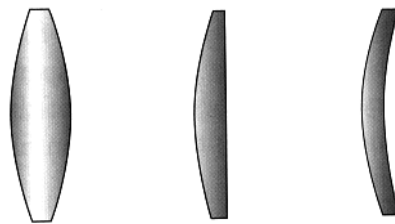
- Diverges parallel rays of light from a point.



- Sometimes called a *diverging lens*.

There are a variety of shapes for each type of lens.

- Convex - thicker in the middle.

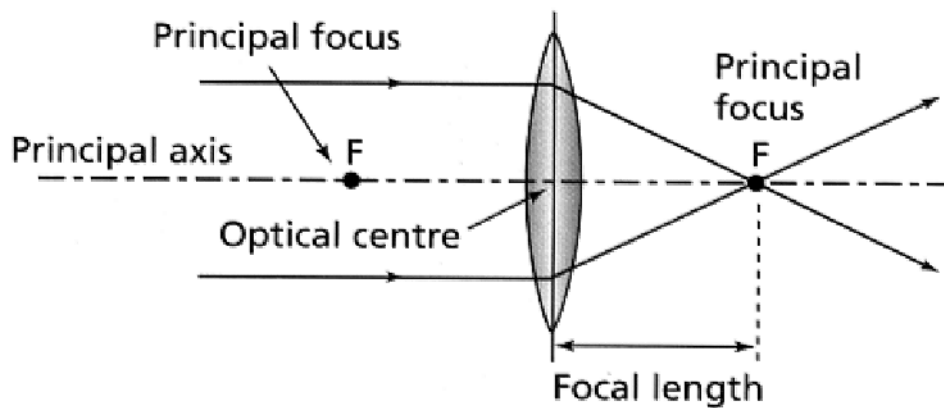


- Concave - thinner in the middle.

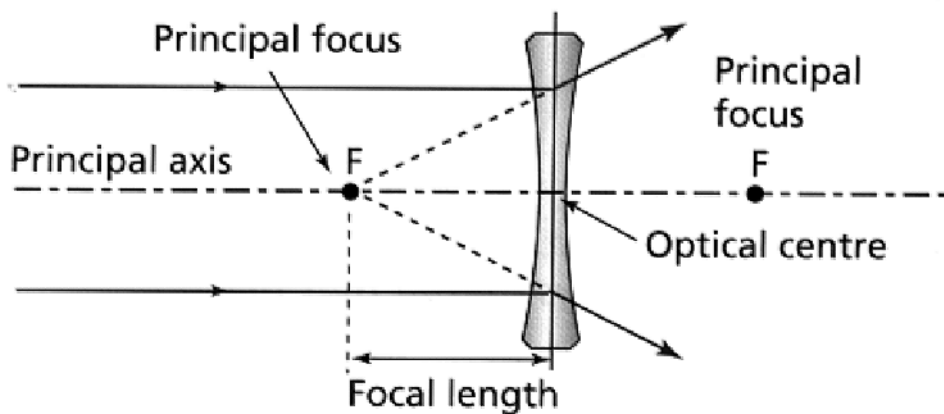


Optical Properties Of Lenses

- Light passing through the optical centre of a lens does not change direction.
- A convex lens **converges** parallel rays of light to a point called the **focus**.
- A concave lens **diverges** parallel rays of light away from the focus.
- The distance between the focus and the centre of the lens is the **focal length**.

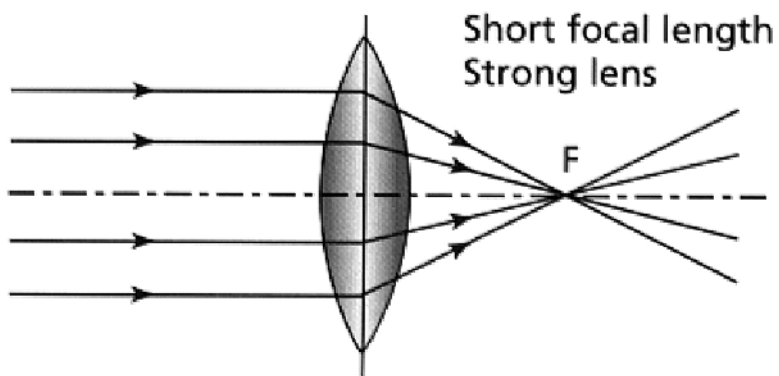
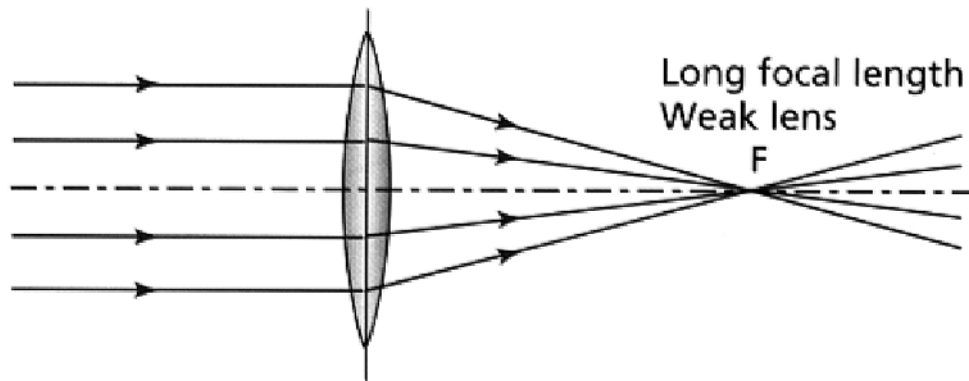


A convex lens.



A concave lens.

- **Powerful lenses** have a **short focal length** and a **larger curvature**.



PRISMS AND LENSES

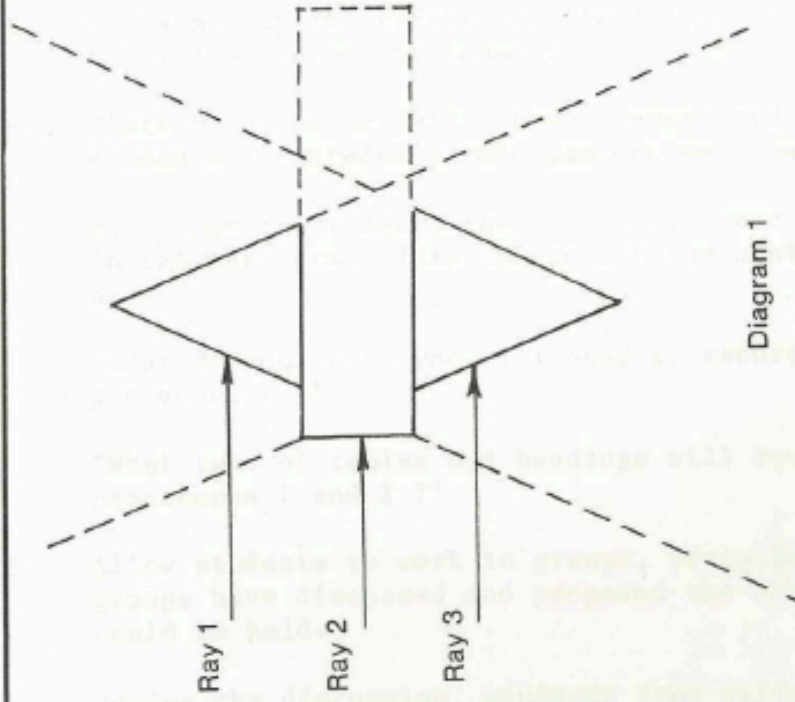


Diagram 1

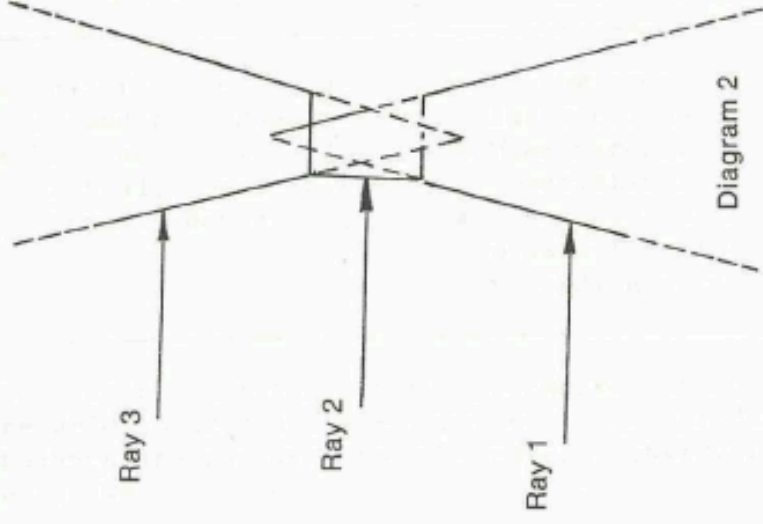


Diagram 2

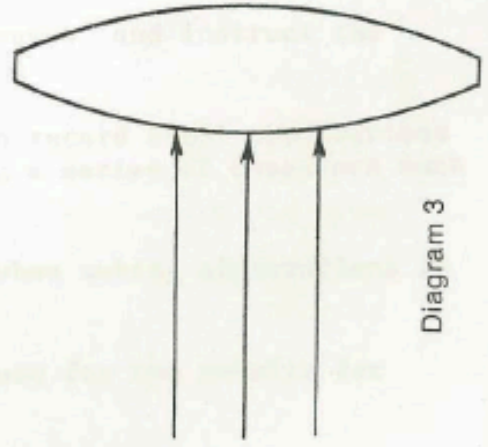


Diagram 3

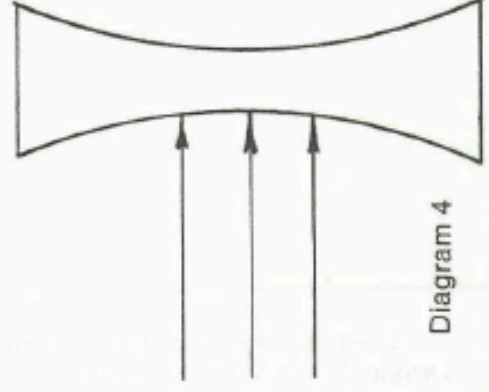


Diagram 4

YEAR 9 SCIENCE PHYSICAL SCIENCES LENSES

- Do the following activity.
- Record your results on a sheet of paper in your group.
- Take a photo and add it below this page.
- Answer the questions.

INVESTIGATION 9.9

Focusing on light

AIM To investigate the transmission of light through different lenses

Materials:

ray box kit

12 V DC power supply

sheet of white paper

ruler and fine pencil

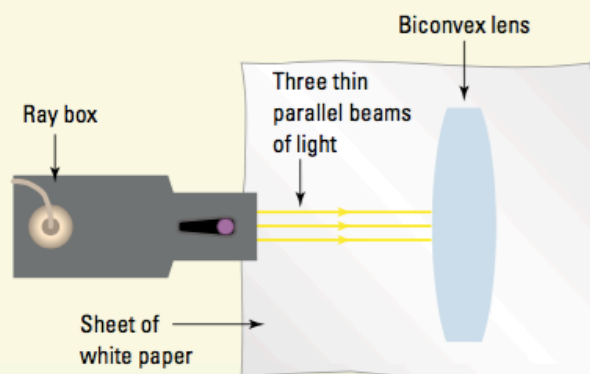
METHOD AND RESULTS

- ▶ Connect the ray box to the power supply and place it on a page of your notebook.

Part A: Biconvex lenses

- ▶ Place the thinner of the two biconvex lenses in the kit on the page and trace out its shape. Project three thin parallel beams of white light towards the lens.

- 1 Trace the paths of the light rays as they enter and emerge from the lens. Remove the lens from the paper so that you can draw the paths of the light rays through the lens.



- ▶ Replace the thin biconvex lens with a thicker one and repeat the previous steps.

Part B: Biconcave lenses

- ▶ Place the thinner of the two biconcave lenses on your notebook page and trace out its shape.

- 2 Trace the path of each of the three thin light beams as they enter and emerge from the lens. Remove the lens from the page so that you can draw the paths of the light beams through the lens.

DISCUSS AND EXPLAIN

- 3 State the focal length (distance from the focal point to the centre of the lens) for each lens.
- 4 Which of the biconvex lenses bends light more, the thin one or the thicker one?
- 5 Explain why the middle light ray does not bend.
- 6 How many times do each of the other rays bend before arriving at the focal point?
- 7 Do the diverging rays come to a focus?
- 8 Do the diverging rays appear to be coming from the same direction? Use dotted lines on your diagram to check.
- 9 Predict where the diverging rays will appear to come from if you use a thicker biconcave lens. Check your prediction with the thicker biconcave lens in the ray box kit.

Observations

- Place your photo of the results here.

Discuss and Explain

- Answer the questions here.

HEATING AND COOLING

Heat and Temperature

- These are **NOT** the same.

Heat: the total of the potential and kinetic energy (internal energy) of the particles in a substance.

Temperature: a measure of the average kinetic energy of the particles in a substance.

- The amount of heat held varies with the type of substance.**

e.g. Metals hold quite a lot of heat, but water holds much more. Hence, it takes more heat to change the temperature of the water than a metal.

Example

A group of students held a large metal nut in a Bunsen flame and heated it to red heat ($\sim 400\text{ }^{\circ}\text{C}$). They then placed it carefully into 50 mL of water at $20\text{ }^{\circ}\text{C}$. The water temperature rose only to $35\text{ }^{\circ}\text{C}$.

Hence the metal may have been much hotter in temperature but it didn't hold much heat as it only changed the water temperature by a small amount.

- Heat always flows from the higher temperature area to the lower temperature area.**

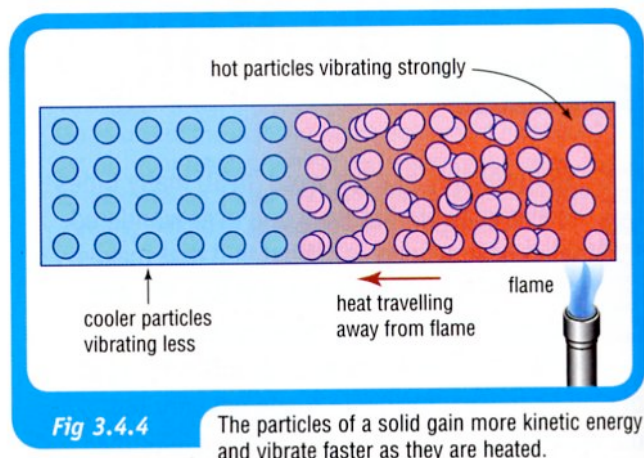
e.g. Your hand is about $35\text{ }^{\circ}\text{C}$ and a bench top is $20\text{ }^{\circ}\text{C}$. When you touch the bench, heat flows from your hand into the bench until both are at the same temperature.

The bench feels cold as the heat is drawn out of your hand.

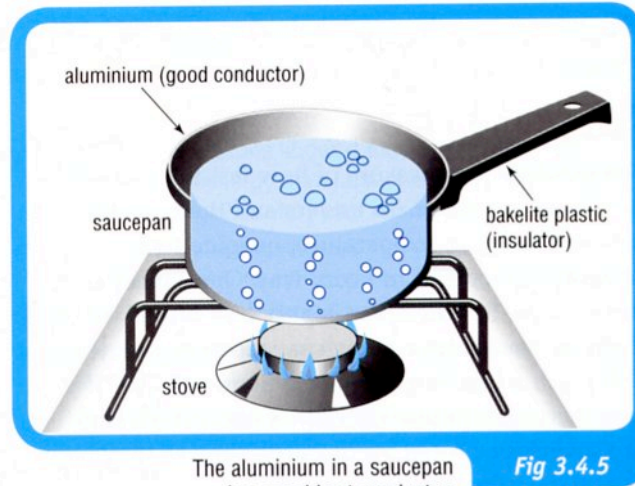
- Heat is transferred in three ways.

Conduction

- Occurs in solids** where particles vibrate in fixed positions.
- Energy is transferred by particles colliding and transferring kinetic energy.
- As the amount of kinetic energy in the particles increases, a temperature rise is felt.



- Some substances are better conductors of heat than others.
- e.g. Metals are very good **conductors**.
Plastics, glass, wood and other similar materials (non-metals) are **insulators** - very poor conductors.

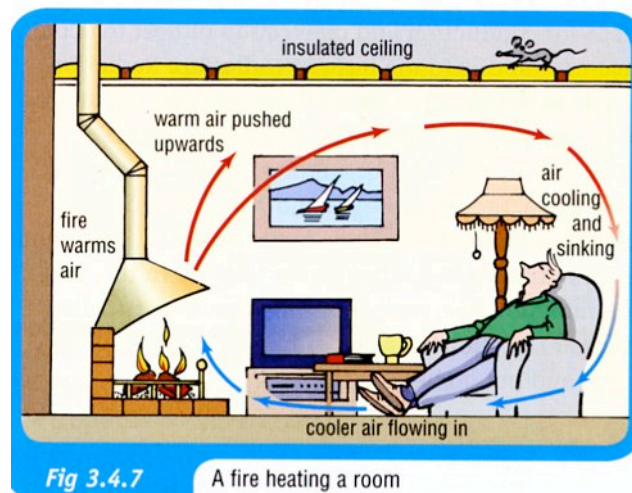


- Liquids and gases are very poor conductors** - their particles are further apart and more mobile so they collide far less.

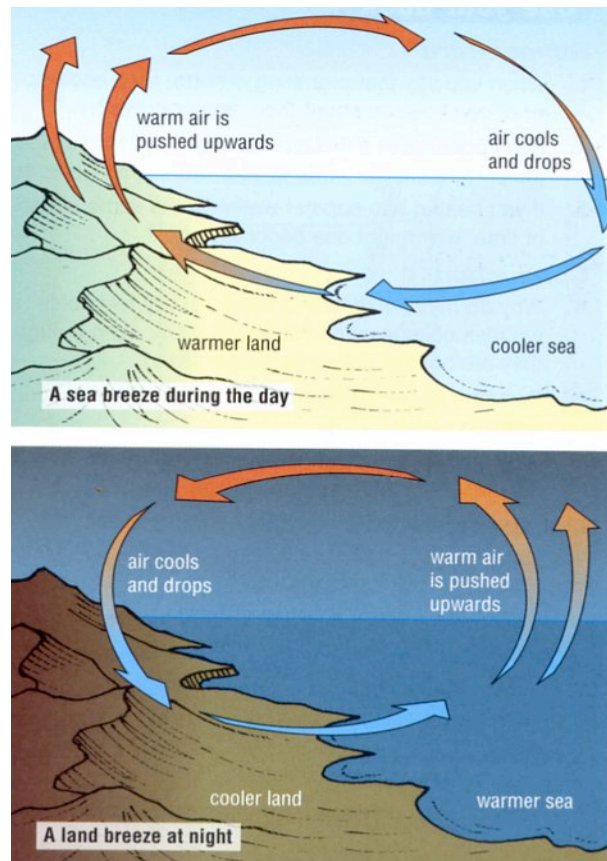
Hence they use another method to transfer heat.

Convection

- Used to transfer heat in liquids and gases.
- Heated particles not only gain kinetic energy but they move further apart, becoming less dense and rising to the top of the liquid or gas.
- Cooler, more dense liquid or gas moves in at the bottom to take the place of the rising particles.
- Hot particles at the top start to cool and become denser, falling back down to the bottom.

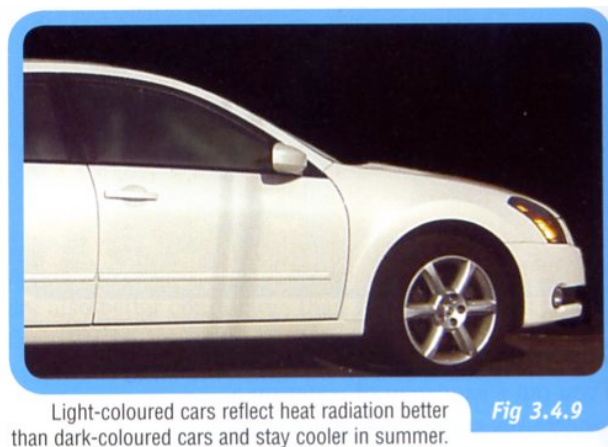


- **Land and sea breezes** are caused by unequal heating of the land and water.
i.e. If the sea is warmer, hot air rises over it and a land breeze moves out to sea.
If the land is warmer, a sea breeze moves in.



Radiation

- Heat is radiated from objects as **infra-red light**.
- Being light, it is able to be transmitted through space or the air at about 300,000 km/sec.
- All objects with heat in them will radiate the heat, and infra-red cameras and film will detect them.
- Infra-red radiation can be reflected, absorbed or transmitted through objects or surfaces that it hits.
- **Black surfaces absorb and radiate heat better than lighter surfaces.**



Investigation - Conduction

Write up the following investigation using the standard Keynote or Pages.

INQUIRY: INVESTIGATION 10.2

Heat conduction in solids

KEY INQUIRY SKILLS:

- processing and analysing data and information
- evaluating

Equipment:

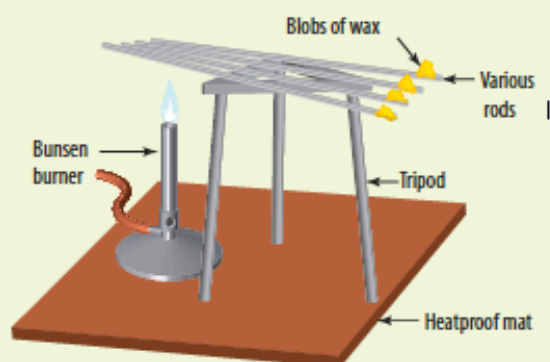
set of three or four metal rods, different metals but identical in size; or heat conduction apparatus

wax candle Bunsen burner and heatproof mat

matches tripod

ruler stopwatch

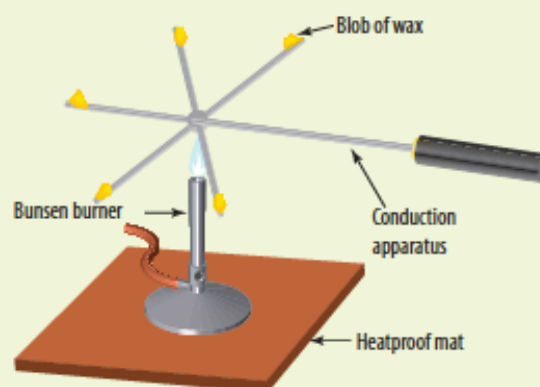
- Set up the tripod and rods or heat conduction apparatus as shown below.
- Light the candle and melt a blob of wax onto one end of each rod. Ensure that each wax blob is the same distance from the end that will be heated by the Bunsen burner flame.



- Use the blue flame of the Bunsen burner to heat the end of each rod. Start the stopwatch at the instant that heating begins. Record the time taken for each blob to produce its first droplet of wax.
- Present your data in a table and draw a bar or column graph in which to display them.

DISCUSS AND EXPLAIN

- 1 According to your data, which of the metals is the best conductor of heat?
- 2 According to your data, which of the metals is the poorest conductor of heat?
- 3 Compare your data with that of others in your class. Comment on the consistency of the conclusions within your class. If there was inconsistency, suggest one or more reasons for it.



Investigation - Absorption and Radiation of Heat

Write up the following investigation using the standard Keynote or Pages.

INQUIRY: INVESTIGATION 10.3

Radiating and absorbing radiant heat

KEY INQUIRY SKILLS:

- questioning and predicting
- processing and analysing data and information

Equipment:

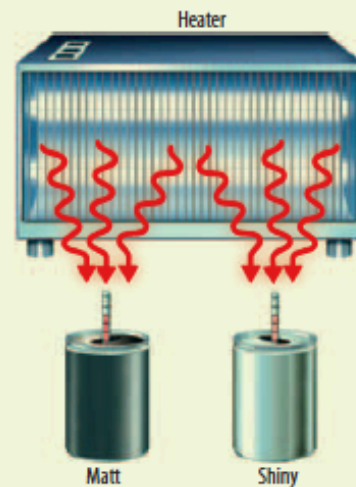
heater or microscope lamp

2 identical shiny, empty, soft-drink cans

matt black paint and paintbrush

2 thermometers or data logger and 2 temperature sensors

- Paint one of the cans matt black and leave the other as it is.
- Construct a table or spreadsheet headed 'Radiating heat' in which to record the temperature every two minutes for up to 14 minutes.
- Make a prediction about your results and write down a hypothesis.
- Pour equal amounts of hot water into each can.
- Place a thermometer or temperature probe into each can. Measure the initial temperature of the hot water and again every two minutes.
- Enter your data into the table or spreadsheet.
- Empty the cans and pour equal amounts of cold tap water into each can.
- Construct a table or spreadsheet headed 'Absorbing radiant heat' in which to record the temperature every two minutes for up to 14 minutes.
- Make a prediction about your results and write down a hypothesis.



- Place a thermometer or temperature probe into each can. Measure the initial temperature of the water.
- Place the two cans at equal distances from a heater or microscope lamp.
- Measure and record your data into the table or spreadsheet every two minutes.
- Plot line graphs that show how the temperature changed over 14 minutes during the cooling and heating of the cans. You may wish to plot both graphs on the same set of axes.

DISCUSS AND EXPLAIN

- 1 Which can radiated heat more quickly?
- 2 Which can absorbed heat more quickly?
- 3 Were each of your hypotheses supported by the data?
- 4 Why was it important to use cans that were identical in size and shape?
- 5 What other variables had to be controlled during this experiment?

Conduction and convection

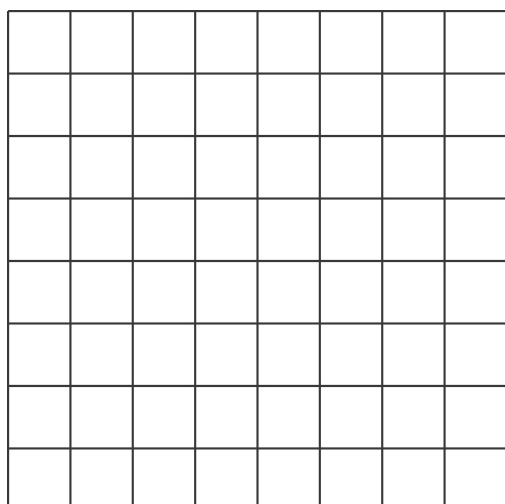
Student: Class:

Testing conduction

1. The heat conduction properties of a new material were tested. One end of a 100-cm length of the material was heated, and the temperature of the rod at the other end was recorded every minute. The data is tabulated below.

(a) Plot a line graph of the data and draw the line of best fit. (Start your graph at 20 °C).

Time (min)	Temperature (°C)
0	20.0
1	25.0
2	29.8
3	35.2
4	40.0
5	44.7
6	50.3
7	54.9
8	59.9



(b) Use your graph to estimate the temperature of the rod at 3.6 minutes after the start.

.....

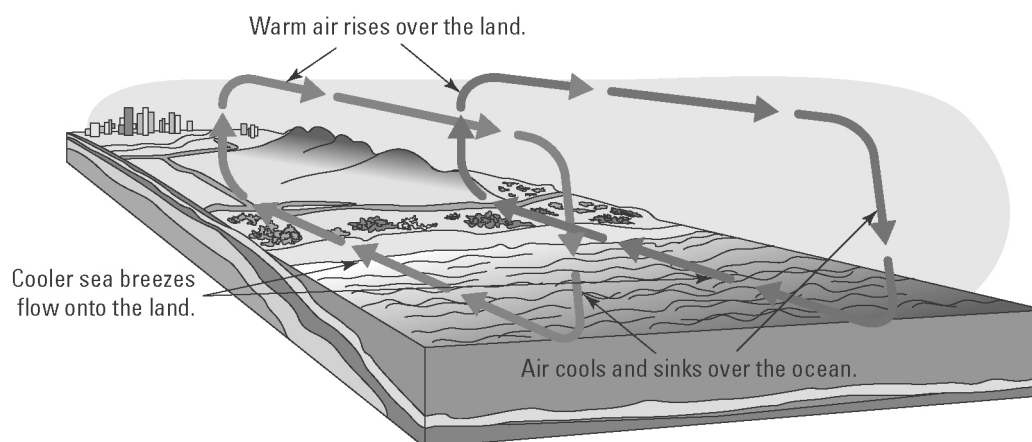
(c) Identify the dependent and independent variables in this experiment.

.....

.....

Convection

2. Consider the following illustration showing the movement of air near the ocean shore. Describe how the production of convection currents makes the climate near the coast milder than inland.



.....

.....

.....

.....

.....

.....

.....

.....

Invisible waves:

Summary

Student: Class:

Use the listed words to complete the sentences.

convex	focal	frequency	microwaves	opaque
plane	rays	reflection	scattered	speed
straight	translucent	transparent	transverse	

1. Light is a form of energy. Light travels in lines. The lines that are used to show the path of light are called
2. Light travels at the of 300 000 kilometres per second in a vacuum.
3. Rays or beams of light may be seen when light is or reflected from particles in substances and redirected to the eye.
4. Light reflected from a mirror follows the law of which states: the angle of incidence equals the angle of reflection. Flat mirrors are called mirrors.
5. Curved mirrors may be concave (curved inwards) or (curved outwards). Light reflected from concave and convex mirrors also follows the law of reflection. Parallel rays of light are reflected to a point in a concave mirror. Convex mirrors reflect light from a wide angle creating an image reduced in size that captures much of the surrounds.
6. Shiny surfaces reflect light but if a surface absorbs light it is said to be
7. If most of the light travels through a material, the material is said to be Window glass is transparent.
8. Some materials allow some light to travel through but also scatter some of the light. Objects are not clearly visible when viewed through these materials. Frosty glass used in bathroom windows is an example. Such materials are said to be
9. Light is an example of an electromagnetic wave. These waves are waves. The distance from crest to crest is the wavelength of a wave.
10. The electromagnetic spectrum is divided into regions from low frequency to high frequency: radio waves;; infra-red waves; visible waves; ultra-violet waves; X-rays; gamma rays.
11. Analogue and digital television, wireless broadband and mobile phone calls can all be transmitted over long distances using high radio waves, also called microwaves.

REVISION

W-15

YEAR 9 SCIENCE PHYSICAL SCIENCES REVISION - TEST 1

1. (a) How do the particles move in a ***transverse wave***? Give an example of this wave.

- (b) How do the particles move in a ***longitudinal wave***? Give an example of this wave.

- (c) Draw a displacement - distance graph to show the following. Make sure you give appropriate scales and labels.

amplitude = 5.00 cm
wavelength = 10.0 cm

2. (a) How would you describe ***light*** - what is it?

- (b) Does light need a medium (substance) to travel in? How do you know?

3. (a) List the parts of the electromagnetic spectrum in order.

(b) Give a use or characteristic for each part.

4. (a) What are the Laws of Reflection?

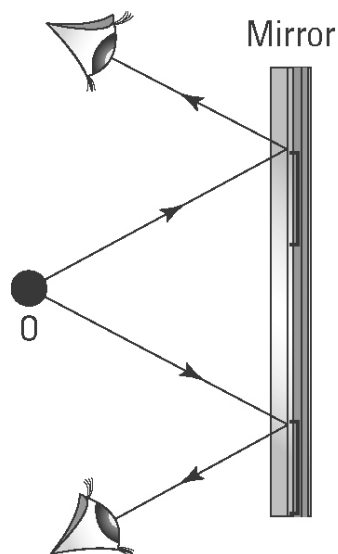
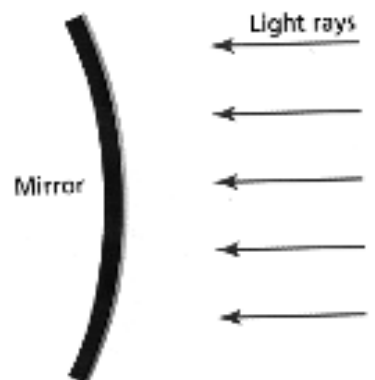
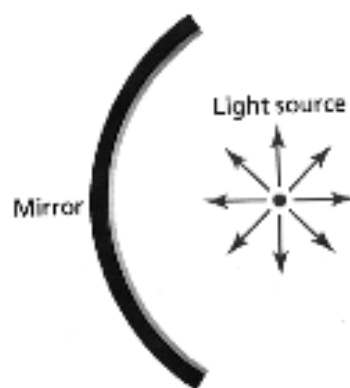
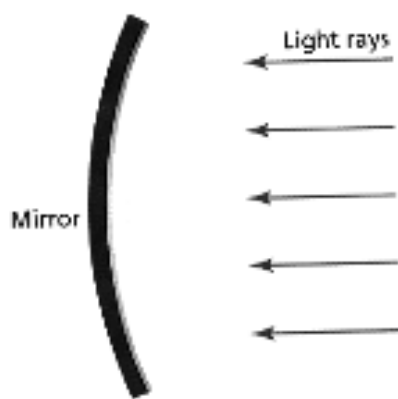
(b) Draw a simple diagram to show it.

5. What is the difference between a **real image** and a **virtual image**?

Virtual image:

Real image:

6. Complete the following ray diagrams.



7. Explained how heat is transferred in the following.
- (a) A piece of copper wire that is heated at one end.
 - (b) A pot of warm water that is being heated.
 - (c) A room with a small gas heater at floor level.
8. **Metals** have a very high thermal conductivity (*able to transfer heat quickly*) while **gases** are very low. Explain this difference.
9. Some very expensive cookware has copper bottoms on all the pots and pans. Suggest a reason for this.

10. Sports people who injure themselves are advised to put ice or a cold pack immediately on the injury to limit any internal bleeding. Why does the ice feel cold on the skin?
11. Radiators on cars, trucks and other engines are painted black and have lots of very fine "fins" which allow the air to pass through. How do these two features make the radiator an efficient cooling device for the water carrying heat away from the engine?
- -
12. Draw a simple labelled diagram to explain how a **sea breeze** occurs during the summer.

13. Parents become quite annoyed when one of their children stands in front of the refrigerator with the door open, thinking about what they would like to take from it. The cold air "falls" out of the fridge onto their feet and the motor starts to chill the inside again once the door closes. This costs money in the long run!

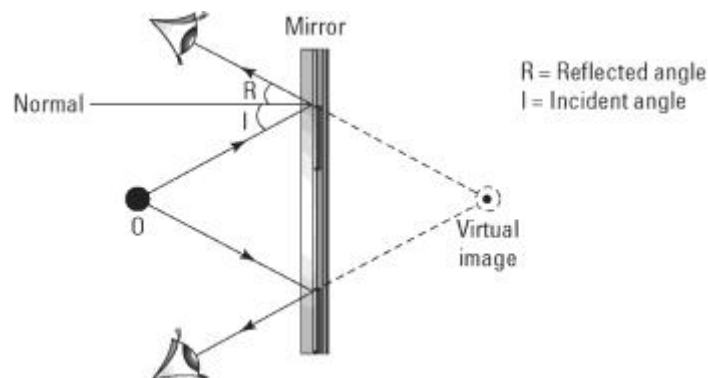
Why does the cold air "fall" out?

Reflection and scattering of light

Answers

Locating the virtual image

1. An object (O) is observed by looking at the rays of light reflecting off a plane mirror. The mirror consists of a plate of glass with a silver back surface mounted on a sheet of wood.



- (a) Use a ruler and pencil to extend each reflected ray (using dotted lines) until they intersect behind the mirror. Label the point of intersection as 'virtual image'.
- (b) Complete the sentence:
The object in front of a reflecting surface and a virtual image are the same... distance from a mirror.
- (c) Draw the *normal* to the mirror surface and measure the incident and reflected angles for the ray above the object, O. From above diagram $R = I$

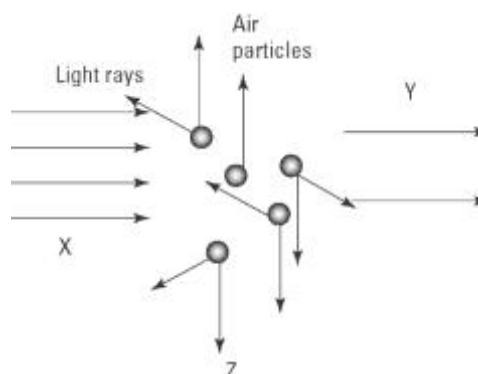
Light and air

2. The diagram below shows the collision of light rays with particles of air in the atmosphere. Use the letters X, Y and Z to identify the following rays.

Scattered:Z.....

Incident:X.....

Transmitted:Y.....

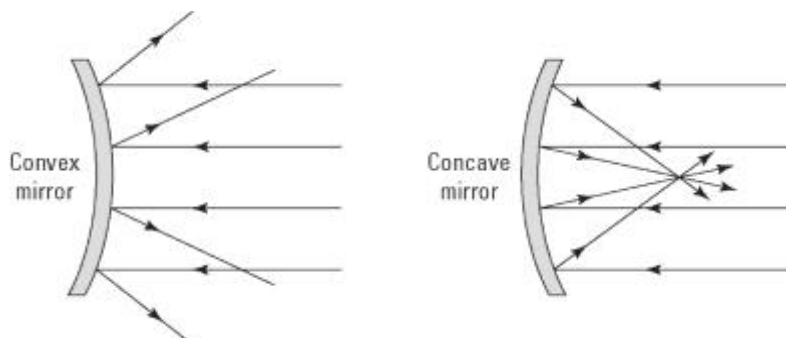


Curved mirrors

Answers

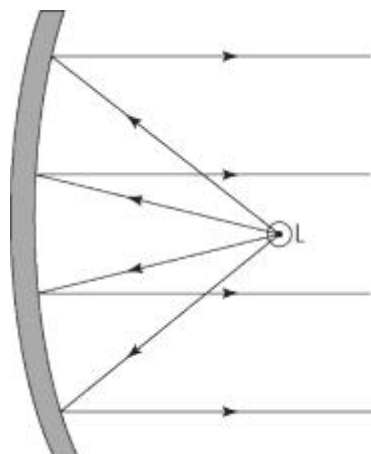
Reflection from convex and concave mirrors

- Four parallel light rays are incident on a convex and a concave mirror as shown. Complete the diagrams (using a pencil and ruler) to show the reflected rays.



Light emitted from the focal point

- The following diagram shows four light beams emitted from a light globe (L) placed at the focal point of the concave mirror.
 - Use a pencil and ruler to show the direction of the light rays reflected from the mirror.



- Describe these reflected rays.
The reflected rays are straight and parallel.

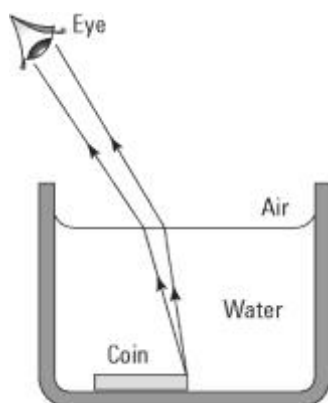
.....

Refraction

Answers

The appearing coin

1. A coin is placed in an aluminium beaker so that it is just out of sight when looking into the beaker from an angle, as shown. Water is then added to the beaker, so that when the water reaches a certain depth, the edge of the coin can be seen.

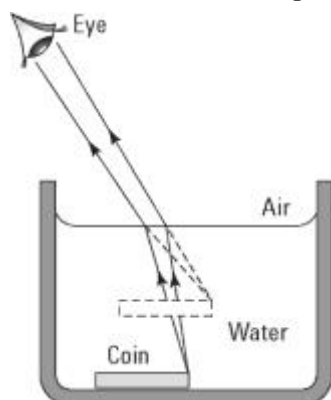


- (a) Use a ruler and pencil to show how the two light rays from the edge of the coin will enter the eye so that the edge becomes visible to the person when the water is added.

- (b) Explain why these rays bend as they enter the air.

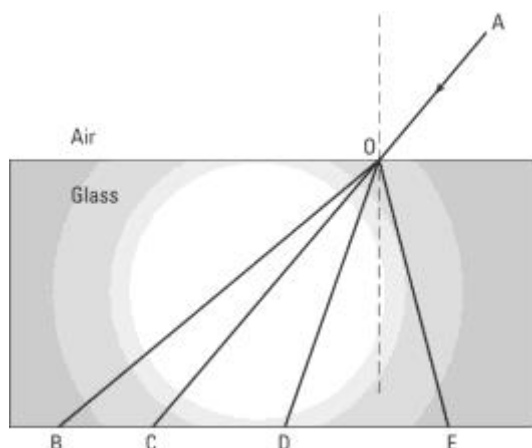
The air has a lower refractive index than water and the light travels at a greater speed in air than in water and so the rays bend when they leave the water.

- (c) On the diagram sketch the apparent position of the coin and show why the coin appears to be at a more shallow depth.



Refraction of light through a rectangular prism

2. A light ray is incident on one side of a glass rectangular prism as shown.



- (a) The light ray travels from A to O. Identify which of the following paths it will take on entering the glass prism.

- A. OB
- B. OC
- C. OD
- D. OE

Answer:

OD

- (b) Use a pencil and ruler to show the path of the light ray as it emerges back into the air at the base of the prism. See above diagram.

Invisible waves: Summary

Answers

Use the listed words to complete the sentences.

convex	focal	frequency	microwaves	opaque
plane	rays	reflection	scattered	speed
straight	translucent	transparent	transverse	

1. Light is a form of energy. Light travels in **straight** lines. The lines that are used to show the path of light are called **rays**
2. Light travels at the **speed** of 300 000 kilometres per second in a vacuum.
3. Rays or beams of light may be seen when light is **scattered** or reflected from particles in substances and redirected to the eye.
4. Light reflected from a mirror follows the law of **reflection** which states: the angle of incidence equals the angle of reflection. Flat mirrors are called **plane** mirrors.
5. Curved mirrors may be concave (curved inwards) or **convex** (curved outwards). Light reflected from concave and convex mirrors also follows the law of reflection. Parallel rays of light are reflected to a **focal** point in a concave mirror. Convex mirrors reflect light from a wide angle creating an image reduced in size that captures much of the surrounds.
6. Shiny surfaces reflect light but if a surface absorbs light it is said to be **opaque**
7. If most of the light travels through a material, the material is said to be **transparent** Window glass is transparent.
8. Some materials allow some light to travel through but also scatter some of the light. Objects are not clearly visible when viewed through these materials. Frosty glass used in bathroom windows is an example. Such materials are said to be **translucent**
9. Light is an example of an electromagnetic wave. These waves are **transverse** waves. The distance from crest to crest is the wavelength of a wave.
10. The electromagnetic spectrum is divided into regions from low frequency to high frequency: radio waves; **microwaves**; infra-red waves; visible waves; ultra-violet waves; X-rays; gamma rays.
11. Analogue and digital television, wireless broadband and mobile phone calls can all be transmitted over long distances using high **frequency** radio waves, also called microwaves.

Conduction and convection

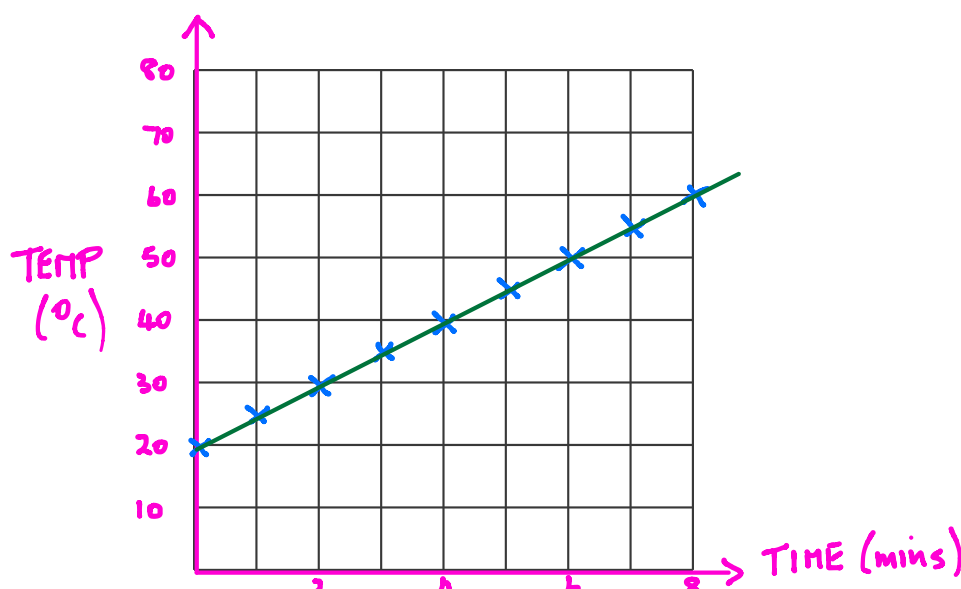
Student: Class:

Testing conduction

1. The heat conduction properties of a new material were tested. One end of a 100-cm length of the material was heated, and the temperature of the rod at the other end was recorded every minute. The data is tabulated below.

(a) Plot a line graph of the data and draw the line of best fit. (Start your graph at 20 °C).

Time (min)	Temperature (°C)
0	20.0
1	25.0
2	29.8
3	35.2
4	40.0
5	44.7
6	50.3
7	54.9
8	59.9



- (b) Use your graph to estimate the temperature of the rod at 3.6 minutes after the start.

38 °C

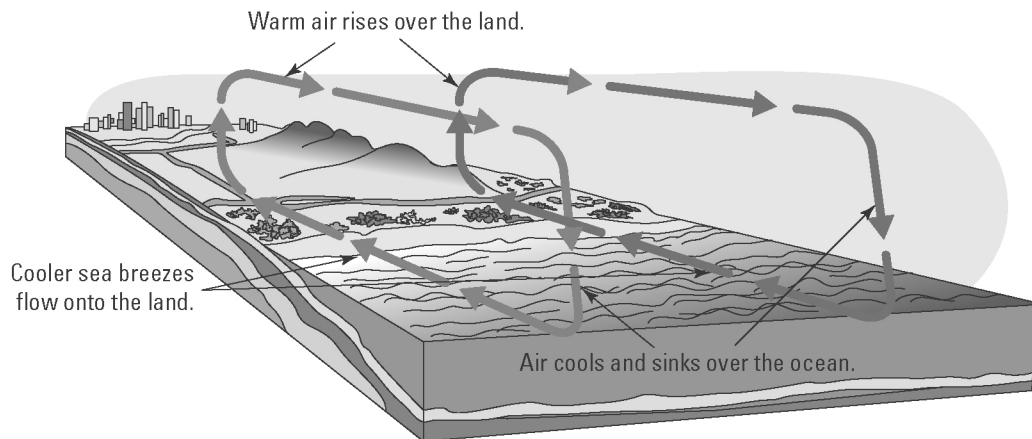
- (c) Identify the dependent and independent variables in this experiment.

Dependent - temperature

Independent - time

Convection

2. Consider the following illustration showing the movement of air near the ocean shore. Describe how the production of convection currents makes the climate near the coast milder than inland.



- Warm air rises above land and moves out to sea.
- Cool air from the sea moves in over the land to take the place of the rising warm air.
- The "sea breeze" cools everything down.

ANSWERS

W-15

YEAR 9 SCIENCE PHYSICAL SCIENCES REVISION - TEST 1

1. (a) How do the particles move in a **transverse wave**? Give an example of this wave.

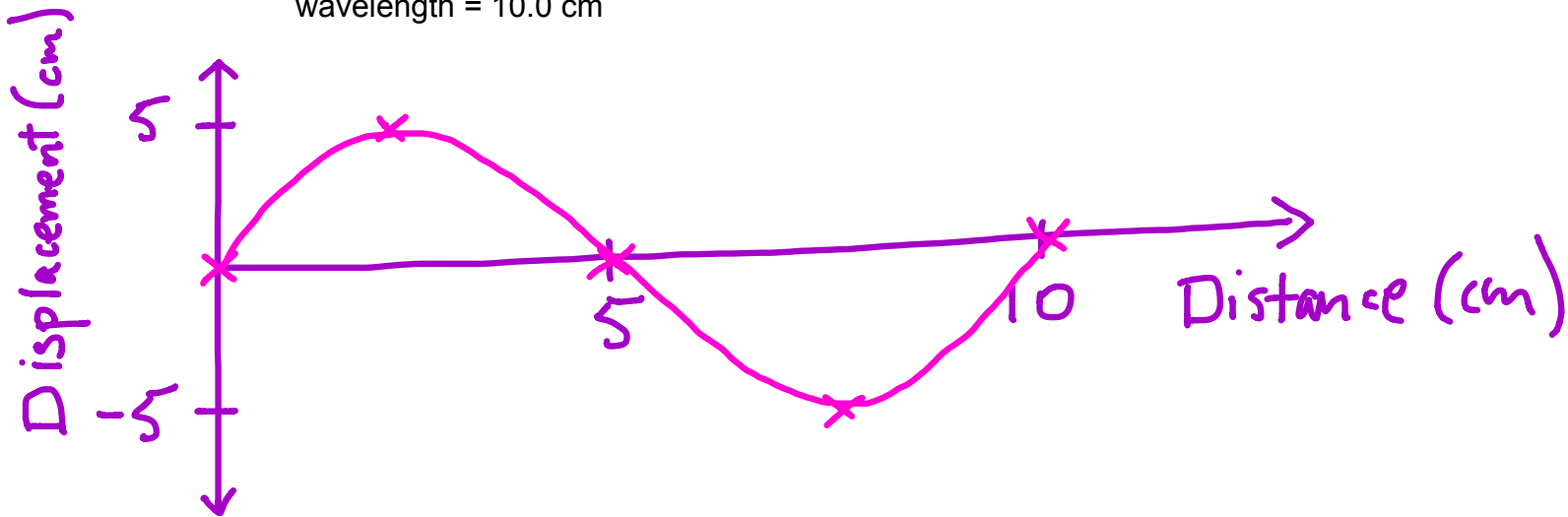
Particles move up + down - at right angles to the direction of the wave.
e.g. water waves, light

- (b) How do the particles move in a **longitudinal wave**? Give an example of this wave.

Particles move backwards + forwards - in same direction as direction of the wave.
e.g. sound

- (c) Draw a displacement - distance graph to show the following. Make sure you give appropriate scales and labels.

amplitude = 5.00 cm
wavelength = 10.0 cm



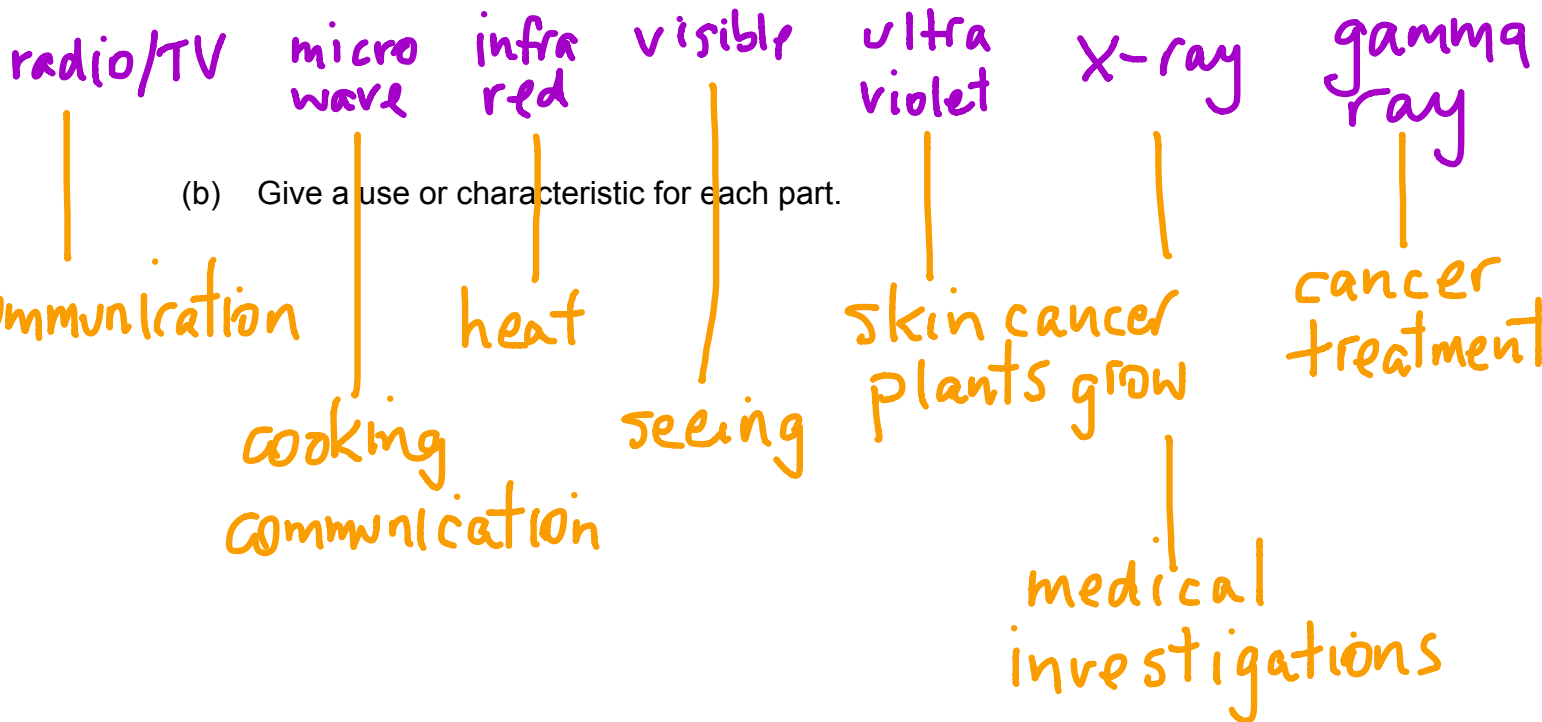
2. (a) How would you describe **light** - what is it?

Pure energy - electromagnetic radiation.

- (b) Does light need a medium (substance) to travel in? How do you know?

No - light travels in space (a vacuum) from the Sun and stars.

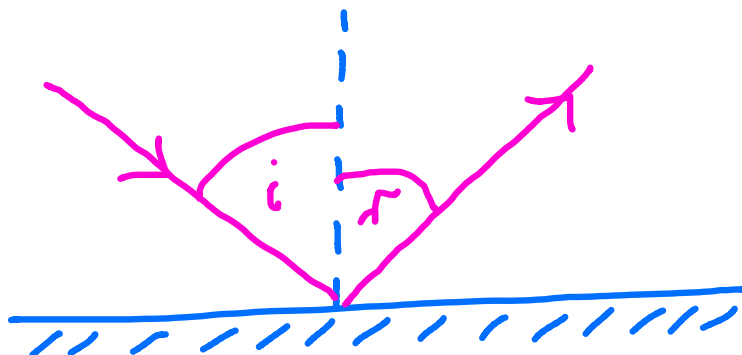
3. (a) List the parts of the electromagnetic spectrum in order.



4. (a) What are the Laws of Reflection?

Angle of incidence = Angle of reflection

- (b) Draw a simple diagram to show it.



5. What is the difference between a **real image** and a **virtual image**?

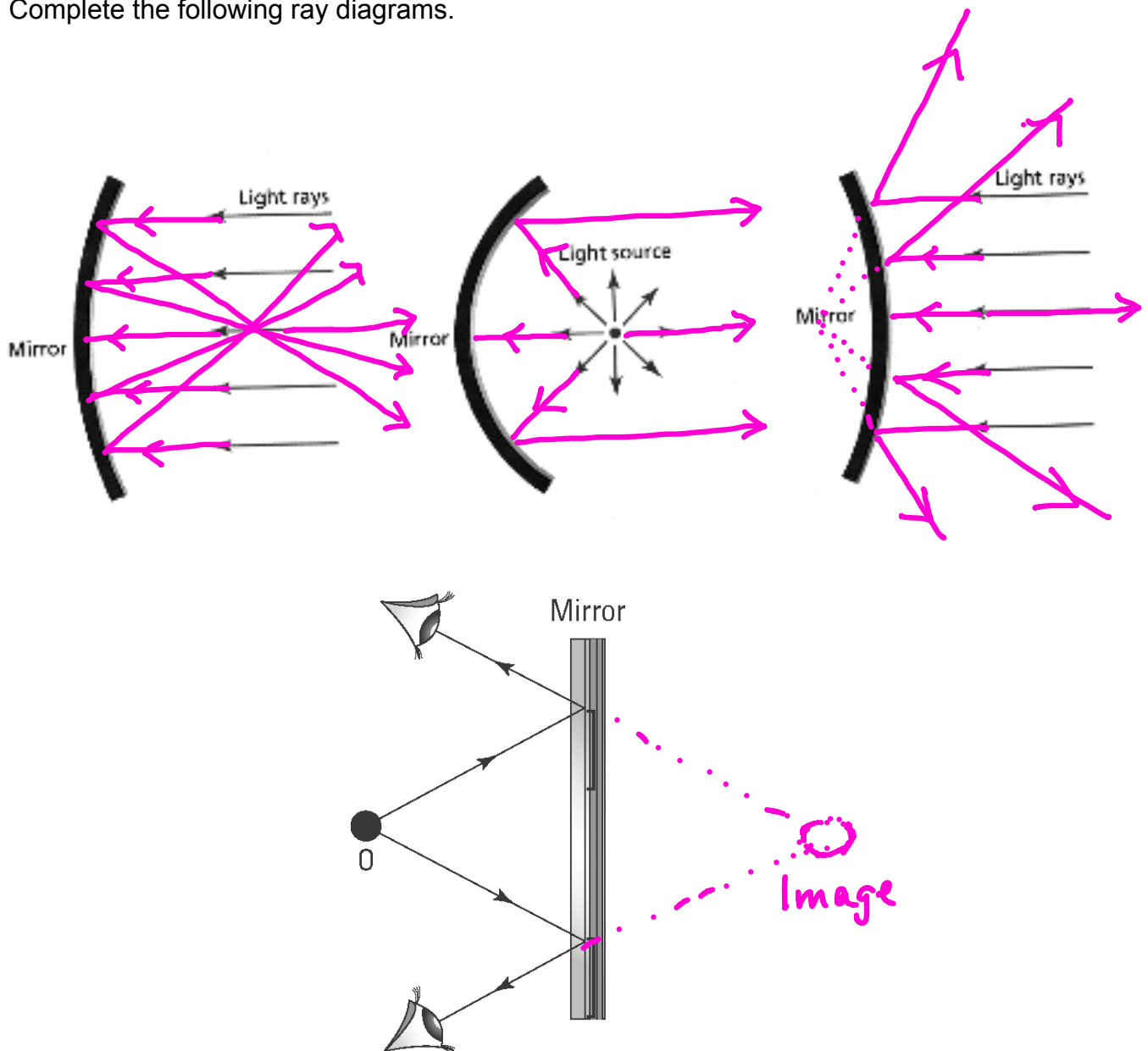
Virtual image:

- Can't be shown on a screen.
- Rays diverge (move away) from each other.

Real image:

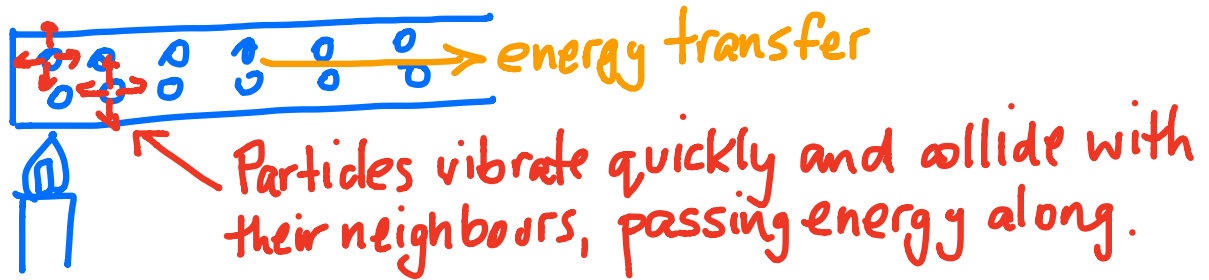
- Can be shown on a screen.
- Ray converge (come together) and cross each other.

6. Complete the following ray diagrams.

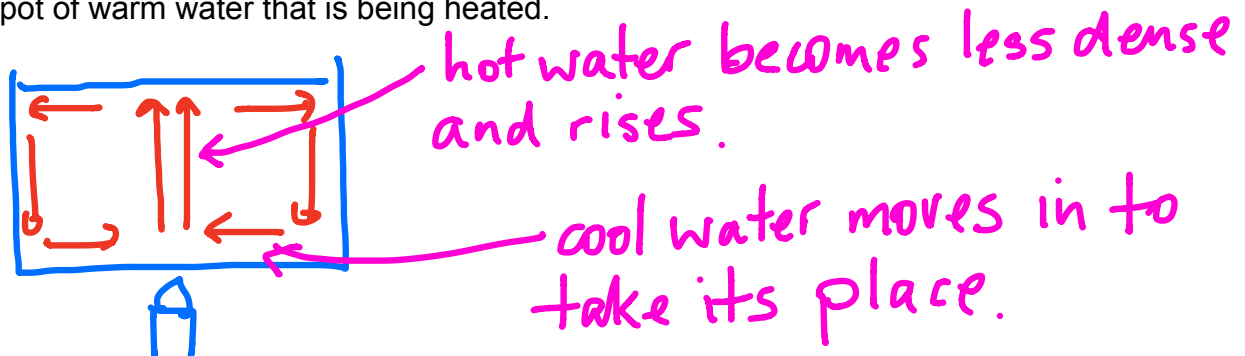


7. Explained how heat is transferred in the following.

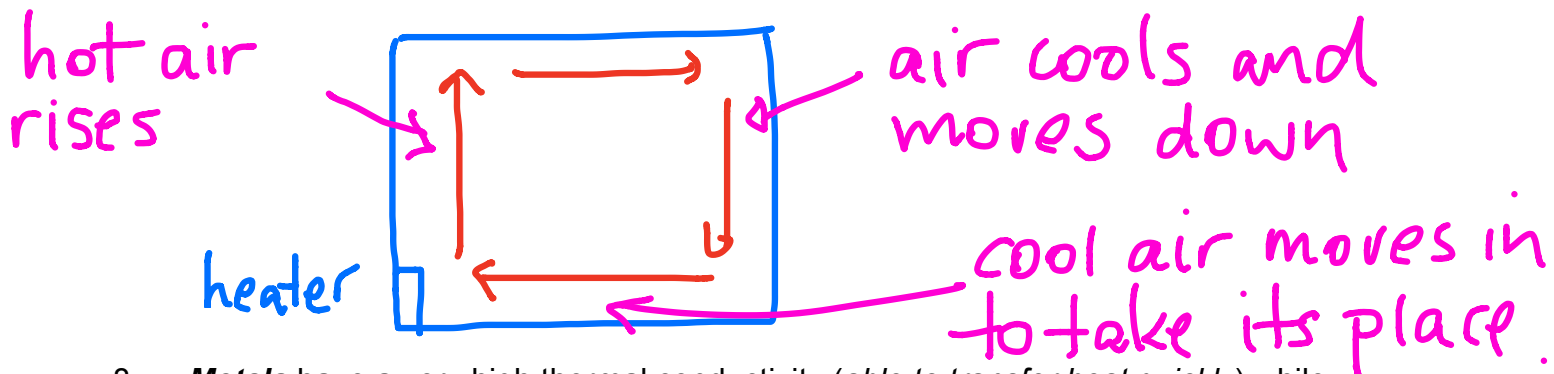
(a) A piece of copper wire that is heated at one end.



(b) A pot of warm water that is being heated.



(c) A room with a small gas heater at floor level.



8. **Metals** have a very high thermal conductivity (*able to transfer heat quickly*) while **gases** are very low. Explain this difference.

Metals- particles close together. Small movement before colliding with the next particle.

Gases- particles very far apart so collisions do not occur easily.

9. Some very expensive cookware has copper bottoms on all the pots and pans. Suggest a reason for this.

Copper is an excellent conductor of heat.
Contents of the pot will heat faster.

10. Sports people who injure themselves are advised to put ice or a cold pack immediately on the injury to limit any internal bleeding. Why does the ice feel cold on the skin?

Heat moves from the skin (hot) to the ice (cold).

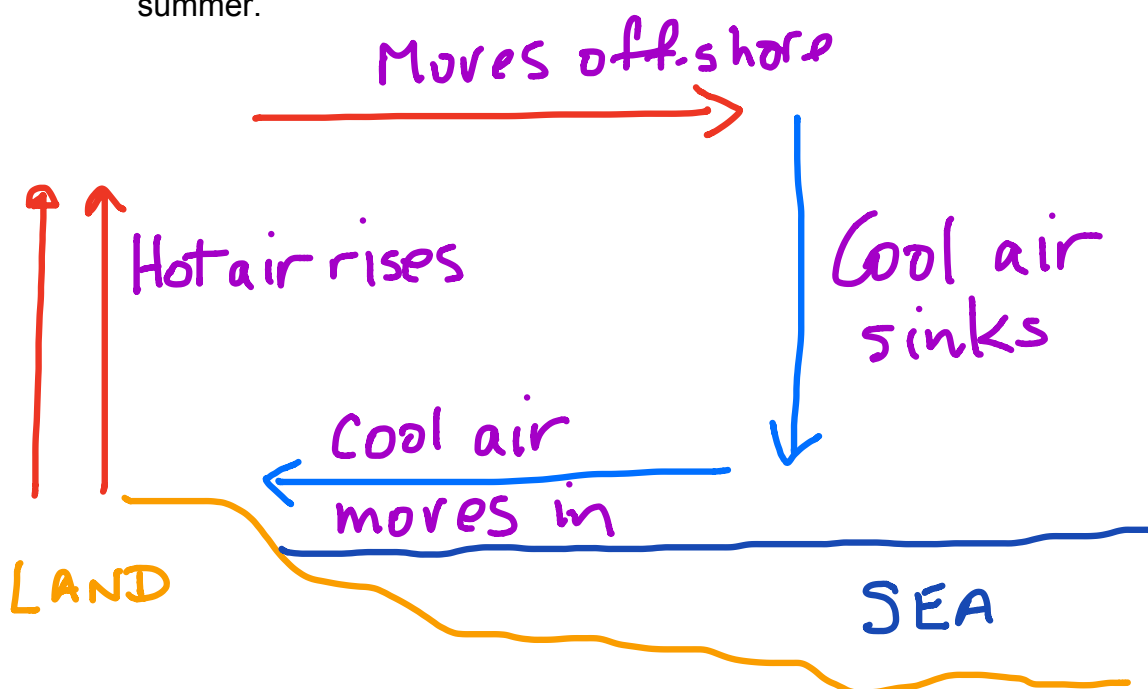
Skin goes cold.

11. Radiators on cars, trucks and other engines are painted black and have lots of very fine "fins" which allow the air to pass through. How do these two features make the radiator an efficient cooling device for the water carrying heat away from the engine?

- Black- radiates heat well.

- Fins - greater surface area allows more heat to leave the system.

12. Draw a simple labelled diagram to explain how a **sea breeze** occurs during the summer.



13. Parents become quite annoyed when one of their children stands in front of the refrigerator with the door open, thinking about what they would like to take from it. The cold air "falls" out of the fridge onto their feet and the motor starts to chill the inside again once the door closes. This costs money in the long run!

Why does the cold air "fall" out?

- Cold air is more dense than warm air.
- It sinks to the lowest level - the floor.